
EXTENDING THE IDEA OF BAYESIAN GROUPED REGRESSION UNDER LOGISTIC SETUP

Submitted by

Tathastu Ghosh

Roll No.:C91/STS/241037

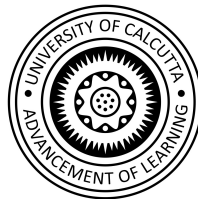
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Supervisors

Prof. Bhaswati Ganguli

Prof. Gaurangadeb Chattopadhyay

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UNIVERSITY OF CALCUTTA
Department of Statistics

Declaration

I, Tathastu Ghosh, student of M.Sc in Statistics at University of Calcutta with registration no. 146-1111-0502-21 and roll no. C91/STS/241037, solemnly declare that the dissertation titled

Extending the Idea of Bayesian Grouped Regression under Logistic Setup

is a genuine and original work undertaken by me under the supervision of Prof. Bhaswati Ganguli and Prof. Gaurangadeb Chattopadhyay, submitted in partial fulfillment of the requirements for the degree of Master of Science in Statistics.

I further affirm that:

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- The project has not been submitted to any other academic degree or assessment at any institution.
- I understand the academic integrity requirements of my university, and I have adhered to all the guidelines and regulations regarding plagiarism and research ethics.
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Abstract

This dissertation extends bayesian grouped regression to the logistic regression setup. The absence of conjugacy in the logistic likelihood is addressed through a data augmentation technique, which enables efficient posterior inference via Gibbs sampling. Three continuous shrinkage priors—the Bayesian Group Horseshoe, GIGG, and GRASP—are empirically evaluated across four simulation scenarios. Their performance is assessed in terms of regression coefficient estimation accuracy, as well as variable selection ability using ROC curves. The proposed methodology is further illustrated through an application to a real-world dataset. The results demonstrate the effectiveness of adaptive shrinkage priors in grouped logistic regression and provide practical insights into their comparative performance and suitability for different modeling settings.

Keywords : Bayesian regression, grouped regression, Logistic, Pólya–Gamma augmentation, Gibbs, Metropolis–Hastings, horseshoe, GRASP, GIGG

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List of Abbreviations

ACF	Auto Correlation Function
AUC	Area under the Curve
BGHS	Bayesian Grouped Horseshoe
ESS	Effective Sample Size
GIGG	Group Inverse-Gamma Gamma
GRASP	Grouped Regression with Adaptive Shrinkage Priors
HPDCI	Highest Posterior Density Credible Interval
MCMC	Markov Chain Monte Carlo
MH	Metropolis-Hastings
MHwG	Metropolis-Hastings within Gibbs
MSE	Mean Square Error
PSRF	Potential Scale Reduction Factor
ROC	Receiver Operating Characteristic

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1 Introduction

The rapid growth of high-dimensional data across science, engineering, medicine and economics has transformed the way statistical models are developed and applied. As a cornerstone of statistical modelling, regression analysis has evolved beyond classical estimation to accommodate high-dimensionality, sparsity, and complex dependence structures commonly encountered in contemporary data.

Group structures are common in regression analysis, they can appear in the form of categorical predictors represented by groups of dummy variables or in the context of additive modelling, where each predictor can be expressed as a set of basis functions forming a group; in applications such as gene expression analysis and financial market modelling, grouping exist naturally in the data. For instance, genes that influence similar traits form groups in gene expression data, while stocks from the same sector form groups in financial data. Leveraging such grouping information is not merely a convenience—it is a statistical necessity for stabilising estimation, reducing overfitting, and improving interpretability in high-dimensional settings.

Bayesian regression provides a coherent framework for incorporating prior information, quantifying uncertainty, and performing regularized estimation in high-dimensional settings. The connection between penalized methods and bayesian inference is well established through the bayesian interpretations of Ridge, Lasso [14] and Elastic Net [11] regression, corresponding to Gaussian and Laplace-type priors or a combination of that on the regression coefficients. Building on these ideas, global–local shrinkage priors were introduced to enable adaptive regularisation, allowing strong shrinkage of negligible coefficients while preserving substantial signals. The horseshoe prior [6] has become a canonical example of this class, and has inspired several extensions aimed at improving flexibility and control over sparsity [18, 20]. Ever since the publication of the horseshoe prior, there has been an explosion of continuous shrinkage priors designed for sparse estimation problems. These priors have become increasingly popular for sparse regression problems because of their good theoretical and empirical properties, in addition to their scale mixture representation, which facilitates straightforward and efficient posterior simulation algorithms. In this direction, more recent proposals such as the Dirichlet–Laplace [3] and normal–beta–prime [2] priors further enrich the class of continuous shrinkage models.

More recently, structured extensions of global–local shrinkage have been developed to incorporate additional hierarchy in the form of grouped predictors. In particular, the *Group Inverse-Gamma Gamma* (GIGG) prior [5] introduces explicit group-level shrinkage through separate hierarchical scales, while the *Grouped Regression with Adaptive Shrinkage Priors* (GRASP) [19] framework allows both local and group-level sparsity to be learned adaptively via beta-prime priors. Xu et al. [22] deploys *Bayesian Grouped Horseshoe prior* (BGHS) which extends the idea of horseshoe in a group setup and its application to additive models.

However, when the response variable is binary— as in disease classification, customer churn prediction, or credit default assessment — *logistic regression* remains the default choice. Bayesian logistic regression inherits the advantages of the bayesian framework, but it presents a fundamental challenge :

the logistic likelihood lacks conjugacy.

It makes the posterior inference analytically intractable and computationally demanding.

Although several computationally intensive approaches have been proposed to address this challenge, this project adopts a data augmentation framework owing to its computational efficiency and compatibility with Gibbs sampling. Specifically, this work extends three prominent grouped Bayesian shrinkage priors—BGHS, GIGG, and GRASP—to the logistic regression setting. Full conditional posterior distributions are derived for each model, enabling the development of efficient Gibbs sampling algorithms. The proposed methods are then evaluated through an extensive simulation study to compare their estimation and variable selection performance. Finally, the methodology is applied to a real-world grouped dataset to demonstrate its practical applicability.

2 Objective

The central aim of this project is to empirically compare three grouped shrinkage priors—BGHS, GIGG and GRASP, under the logistic regression framework. To achieve this, the work is structured around the following specific objectives :

- Extending the idea of the above priors under logistic setup
- Deriving the full conditional posterior distributions for each prior choices
- Deploying the Gibbs sampler and applying these three under several experimental settings
- Empirically compare from the perspective of estimation and prediction accuracy across several replications
- Applying three priors on a real life grouped data.

3 Preliminaries

3.1 Logistic Regression

Many categorical response variables have only two categories: for example whether you take public transport today (yes/no) or whether one can pass the exam or not (yes/no). Here, the response y has two categories denoted often by 1 ("success") or 0 ("failure").

The distribution of y is specified by probabilities $P(y = 1) = \pi$ of success and $P(y = 0) = 1 - \pi$ of failure. Having y , we have multiple explanatory variables denoted by the vector $\mathbf{x}$. The value of π will vary as the $\mathbf{x}$ changes and we replace π by $\pi(\mathbf{x})$ to describe dependence. A linear fit between $\pi(\mathbf{x})$ and $\mathbf{x}$ is surely not plausible here for several reasons. In logistic regression, we use the *logistic function*,

$$\pi(\mathbf{x}) = \frac{e^{\boldsymbol{\beta}^\top \mathbf{x}}}{1 + e^{\boldsymbol{\beta}^\top \mathbf{x}}} \quad (3.1.1)$$

The equation (3.1.1) can be written as (assuming only one predictor)

$$\log \left(\frac{\pi(x)}{1 - \pi(x)} \right) = \beta_0 + \beta_1 x \quad (3.1.2)$$

The left hand side of equation (3.1.2) is called the *log-odds* or *logit*. Hence, logistic regression model has a logit that is linear in x .

Interpretations of coefficients : The intercept β_0 represents the log-odds of $y = 1$ with all predictors fixed at 0 and β_1 represents the change in log-odds of $y = 1$ per unit change in x keeping other variables fixed (if exists).

3.2 Model framework and bayesian point of view

For a binary response variable $y_i \in \{0, 1\}$, we model the probability of success as a function of covariates via the logistic regression model :

$$y_i \mid \mathbf{x}_i \stackrel{\text{indep}}{\sim} \text{Bernoulli}(\pi_i) \quad i = 1, 2, \dots, n$$

and

$$\pi_i = P(y_i = 1 \mid \mathbf{x}_i) = \frac{e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}}, \quad i = 1, 2, \dots, n \quad (3.2.1)$$

where,

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{x}_i = \begin{pmatrix} 1 \\ x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix}$$

In *bayesian regression*, inference is drawn based on the posterior samples of β . The joint posterior distribution of β given $\mathbf{y}, X$ is given by

$$p(\beta \mid \mathbf{y}, X) \propto p(\mathbf{y} \mid X, \beta) \cdot p(\beta) \quad (3.2.2)$$

where, $p(\beta)$ denotes the prior distribution, encapsulating pre-existing beliefs about the parameters and $p(\mathbf{y} \mid X, \beta)$ is the likelihood based on the assumed error structure.

Inference is drawn directly from the posterior samples, yielding point estimates (for example, posterior means) and interval estimates (HPDCI). This framework naturally accommodates prior information, provides coherent uncertainty quantification, and extends readily to hierarchical or regularized specifications.

3.3 Grouped regression framework

By grouped regression, we refer to regression models in which the predictors are naturally partitioned into distinct groups. Such grouping may arise from categorical variables represented by sets of dummy variables, basis function expansions in additive models, or inherent structural relationships among predictors. Grouped regressors are commonly encountered in a wide range of applications, including genomics, metabolomics, neuroimaging, environmental health, and finance. For example, genes participating in the same biological pathway, metabolites belonging to a common metabolic pathway, neighbouring voxels in neuroimaging data, environmental contaminants sharing similar chemical or toxicological characteristics, and stocks belonging to the same industrial sector all constitute natural groups.

In such cases, leveraging relevant grouping information to construct grouped multivariate shrinkage profiles may help achieve additional variance reduction beyond comparable methods that ignore the grouping structure.

3.3.1 Group shrinkage and local shrinkage

Grouped bayesian shrinkage methods employ a hierarchical regularisation mechanism that operates at multiple levels. At the group level, a **group shrinkage parameter** is assigned to each group of predictors and determines the overall importance of that group. Groups with little predictive contribution are collectively shrunk towards zero, effectively excluding them from the model, while groups containing relevant information experience relatively little shrinkage. This induces *group-wise sparsity*, allowing the model to identify the important groups among a large collection of candidates.

Within each group, **local shrinkage parameters** are assigned to individual regression coefficients, providing coefficient-specific adaptivity. They strongly shrink negligible effects while allowing influential predictors to retain relatively large coefficients. Consequently, even within an important group, only the truly relevant predictors are selected.

For example, in an air pollution dataset, predictors can be grouped by their source category (e.g., traffic-related pollutants like CO and NO_2 , industrial pollutants like SO_2 and heavy metals, and secondary pollutants like O_3). While the traffic-related group as a whole may be associated with health outcomes, only a few specific predictors within that group, such as NO_2 and $PM_{2.5}$, may have substantial predictive power. The group shrinkage parameter identifies the relevant source group, whereas the local shrinkage parameters identify the influential individual predictors.

3.3.2 Notations

Let, $\mathbf{y}$ is an $n \times 1$ vector of binary responses, X is an $n \times p$ matrix of standardized regressors, where p denotes the number of predictors. Let, $g = 1, 2, \dots, G$ indexes the groups and p_g denotes the number of predictors in the g^{th} group, hence $\sum_{g=1}^G p_g = p$.

Corresponding to the g -th group, let $\beta_g^{p_g \times 1} = (\beta_{g1}, \beta_{g2}, \dots, \beta_{gp_g})^\top$ denote the vector of regression coefficients and $X_g^{n \times p_g}$ denote the associated design matrix. Consequently, the complete regression coefficient vector and design matrix can be expressed as the concatenation of their group-specific components:

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_G \end{pmatrix} \text{ and } X = (X_1 \mid X_2 \mid \dots \mid X_G)$$

Regression Coefficients	β_1	β_2	β_3	β_4	β_5	β_6	$\cdots$	β_{p-2}	β_{p-1}	β_p
Local Shrinkage Parameters	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	$\cdots$	λ_{p-2}	λ_{p-1}	λ_p
Group Shrinkage Parameters	δ_1				δ_2		$\cdots$		δ_G	
Global Shrinkage Parameter	τ									

Figure 1: An illustration on how the group shrinkage, local shrinkage and global shrinkage parameters interact

The figure 1 illustrates the hierarchical shrinkage framework. At the bottom level, there are regression coefficients, which are regularized by *local shrinkage parameters* λ_j . Note that, any $\beta_k, k = 1, 2, \dots, p$ can be defined as β_{gj} for some g and j as each predictor must belong to any one of the G groups, correspondingly, that local shrinkage parameter is denoted by λ_{gj} . Local shrinkage parameter within the same group g share a common group-level shrinkage parameter δ_g . At the top of the hierarchy, the global parameter τ controls the overall shrinkage intensity.

It is already discussed that the inferences are drawn from the posterior samples of the parameters. It is required to have a generic notation for it. Throughout the subsequent derivations, the notation

$$p(\theta \mid \cdot) \text{ or } \theta \mid \cdot \sim$$

is used to denote the **full conditional distribution** of a parameter θ and the “ $\cdot$ ” symbol represents conditioning on all remaining model parameters, latent variables, hyperparameters and the observed data.

3.4 Global–local shrinkage hierarchy

The likelihood of $y_1, y_2, \dots, y_n$ is given by :

$$\begin{aligned} & \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \\ &= \prod_{i=1}^n \left(\frac{\pi_i}{1 - \pi_i} \right)^{y_i} (1 - \pi_i) \end{aligned} \quad (3.4.1)$$

Using (3.2.1), (3.4.1) simplifies to

$$\prod_{i=1}^n \frac{(e^{\mathbf{x}_i^\top \boldsymbol{\beta}})^{y_i}}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}} \quad (3.4.2)$$

Following the idea introduced by Xu et al. [22], the fully bayesian model along with (3.4.2) can be defined as :

$$\begin{aligned} \beta_{gj} \mid \tau^2, \lambda_{gj}^2, \delta_g^2 &\sim \mathcal{N}(0, \tau^2 \delta_g^2 \lambda_{gj}^2) \\ \lambda_{gj}^2 &\sim \pi(\lambda_{gj}^2) d\lambda_{gj}^2 \\ \delta_g^2 &\sim \pi(\delta_g^2) d\delta_g^2 \\ \tau^2 &\sim \pi(\tau^2) d\tau^2 \end{aligned}$$

3.5 Pólya–Gamma data augmentation

3.5.1 Motivation

Bayesian inference for the logistic regression model has long been recognized as a hard problem, due to the analytically inconvenient form of the model’s likelihood function in (3.4.2). In the two decades since the work of Albert and Chib (1993) [1] on the simple latent–variable method of posterior inference for the probit model, there have been many attempts to apply the same missing data strategy to the logit model, but they all involve data–augmentation algorithms that are either approximate, or are significantly more complicated, as they involve multiple layers of latent variables.

Perhaps as a result, the bayesian treatment of the logistic model has not seen widespread adoption by non–statisticians in the way that, for example the bayesian probit model is used extensively in market research and political science. The lack of standard computational approach also makes it more difficult to use the logit link in the kind of complex hierarchical models.

Polson et al. [15] circumvent this problem in their 2013 paper by presenting a new data–augmentation strategy which is more general and can be applied to more broader class of models. This approach appeals to a new family of distributions namely, *Pólya–Gamma distribution*.

3.5.2 Pólya–Gamma distribution

Definition : A random variable X has a Pólya–Gamma distribution with parameters $b > 0$ and $c \in \mathbb{R}$ if

$$X \stackrel{D}{=} \frac{1}{2\pi^2} \sum_{k=1}^{\infty} \frac{g_k}{(k-1/2)^2 + c^2/(4\pi^2)} \quad (3.5.1)$$

where, $g_k \sim \text{Gamma}(b, 1)$ are independent Gamma random variables and $\stackrel{D}{=}$ indicates equality in distribution. Notationally, $X \sim \text{PG}(b, c)$.

Hence, we can think Pólya–Gamma distribution is a special *subset of the class of the weighted convolutions of gamma distribution*.

The $\text{PG}(b, c)$ distribution possesses several important theoretical properties, some of which are briefly discussed in this section. A fundamental aspect of working with this distribution is the ability to simulate PG random variates efficiently.

Polson et al. [15] proposed an elegant simulation strategy by first considering the baseline distribution $\text{PG}(1, 0)$. The more general $\text{PG}(b, 0)$ distribution is then obtained by representing it as an infinite convolution of gamma random variables. Finally, the entire $\text{PG}(b, c)$ family is obtained through an exponential tilting of the $\text{PG}(b, 0)$ density. Specifically,

$$p(\omega \mid b, c) = \frac{\exp\left(-\frac{c^2}{2}\omega\right) p(\omega \mid b, 0)}{\mathbb{E}_{\omega} \left[\exp\left(-\frac{c^2}{2}\omega\right) \right]},$$

where $p(\omega \mid b, 0)$ denotes the density of a $\text{PG}(b, 0)$ random variable.

3.5.3 Data augmentation strategy

This Pólya–Gamma family has been carefully constructed to yield a simple Gibbs sampler for the Bayesian logistic regression model.

Theorem : Let, $p(\omega)$ denote the density of the random variable $\omega \sim \text{PG}(b, 0)$, $b > 0$. Then the following integral identity holds for all $a \in \mathbb{R}$:

$$\frac{(e^{\psi})^a}{(1 + e^{\psi})^b} = 2^{-b} e^{\kappa\psi} \int_0^{\infty} e^{-\omega\psi^2/2} p(\omega) d\omega \quad (3.5.2)$$

where $\kappa = a - b/2$.

Consider, the notation from subsection 3.2 and define $\psi_i = \mathbf{x}_i^{\top} \boldsymbol{\beta}$, $i = 1, 2, \dots, n$, which are the log-odds of success. To derive our Gibbs sampler, we appeal to the above theorem (Equation (3.5.2)) and write the likelihood contribution of observation i as

$$\begin{aligned} L_i(\boldsymbol{\beta}) &= \frac{(e^{\mathbf{x}_i^{\top} \boldsymbol{\beta}})^{y_i}}{1 + e^{\mathbf{x}_i^{\top} \boldsymbol{\beta}}} = \frac{(e^{\psi_i})^{y_i}}{1 + e^{\psi_i}} \\ &= 2^{-1} e^{\kappa_i \psi_i} \int_0^{\infty} e^{-\omega_i \psi_i^2/2} p(\omega_i) d\omega_i \\ &\propto e^{\kappa_i \mathbf{x}_i^{\top} \boldsymbol{\beta}} \int_0^{\infty} e^{-\omega_i (\mathbf{x}_i^{\top} \boldsymbol{\beta})^2/2} p(\omega_i) d\omega_i \end{aligned} \quad (3.5.3)$$

The equation (3.5.3) is obtained by considering $\kappa_i = a_i - b/2$, where $a_i = y_i$ and $b = 1$, hence $p(\omega_i)$ is the density of PG(1, 0).

The representation reveals the joint distribution of y_i and ω_i given β :

$$p(y_i, \omega_i \mid \beta) \propto \exp \left\{ (y_i - 1/2) \mathbf{x}_i^\top \beta \right\} \exp \left\{ -\omega_i (\mathbf{x}_i^\top \beta)^2 / 2 \right\} p(\omega_i) \quad (3.5.4)$$

Note that, when we integrate out ω_i from (3.5.4) then we will go back to $L_i(\beta)$ using the above theorem. When we condition on ω_i — that is, when we treat the latent variable as known, the term $p(\omega_i)$ is constant and can be absorbed into the proportionality constant. Hence,

$$p(y_i \mid \omega_i, \beta) = \frac{p(y_i, \omega_i \mid \beta)}{p(\omega_i \mid \beta)} \propto \exp \left\{ (y_i - 1/2) \mathbf{x}_i^\top \beta \right\} \exp \left\{ -\omega_i (\mathbf{x}_i^\top \beta)^2 / 2 \right\}$$

which is the kernel of a Normal distribution in $\psi_i = \mathbf{x}_i^\top \beta$.

3.5.4 Conjugacy under the augmented model

The difficulty is removed by introducing auxiliary latent variables ω_i — one for each observation— such that the non-linear logistic likelihood becomes Gaussian when conditioned on these latent variables. Consider the conditional likelihood in (3.5.3) and taking a $\mathcal{N}(\mathbf{b}, B)$ prior on β . The posterior of β given $\omega = (\omega_1, \omega_2, \dots, \omega_n)$:

$$\begin{aligned} p(\beta \mid \omega, \mathbf{y}) &\propto p(\beta) \cdot \prod_{i=1}^n p(y_i \mid \omega_i, \beta) = p(\beta) \cdot \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \beta - \omega_i (\mathbf{x}_i^\top \beta)^2 / 2 \right\} \\ &\propto p(\beta) \cdot \prod_{i=1}^n \exp \left\{ -\frac{\omega_i}{2} \left(\mathbf{x}_i^\top \beta - \kappa_i / \omega_i \right)^2 \right\} \\ &\propto p(\beta) \cdot \exp \left\{ -\frac{1}{2} (\mathbf{z} - X\beta)^\top \Omega (\mathbf{z} - X\beta) \right\} \end{aligned} \quad (3.5.5)$$

where, $\mathbf{z} = (\kappa_1 / \omega_1, \kappa_2 / \omega_2, \dots, \kappa_n / \omega_n)$ and $\Omega = \text{diag}(\omega_1, \omega_2, \dots, \omega_n)$.

Consider, the exponent of the term in the expression (3.5.5)

$$\begin{aligned} &(\beta - \mathbf{b})^\top B^{-1} (\beta - \mathbf{b}) + (\mathbf{z} - X\beta)^\top \Omega (\mathbf{z} - X\beta) \\ &= \left(\beta - (X^\top \Omega X + B^{-1})^{-1} (X^\top \Omega \mathbf{z} + B^{-1} \mathbf{b}) \right)^\top (X^\top \Omega X + B^{-1}) \\ &\quad \left(\beta - (X^\top \Omega X + B^{-1})^{-1} (X^\top \Omega \mathbf{z} + B^{-1} \mathbf{b}) \right) + \text{terms not involving } \beta \end{aligned} \quad (3.5.6)$$

Using expression (3.5.6) in (3.5.5),

$$\begin{aligned} p(\beta \mid \omega, \mathbf{y}) &\propto \exp \left\{ -\frac{1}{2} (\beta - \mathbf{b})^\top B^{-1} (\beta - \mathbf{b}) \right\} \exp \left\{ -\frac{1}{2} (\mathbf{z} - X\beta)^\top \Omega (\mathbf{z} - X\beta) \right\} \\ &\propto \exp \left[-\frac{1}{2} \left\{ \left(\beta - (X^\top \Omega X + B^{-1})^{-1} (X^\top \Omega \mathbf{z} + B^{-1} \mathbf{b}) \right)^\top \right. \right. \\ &\quad \left. \left. (X^\top \Omega X + B^{-1}) \left(\beta - (X^\top \Omega X + B^{-1})^{-1} (X^\top \Omega \mathbf{z} + B^{-1} \mathbf{b}) \right) \right\} \right] \end{aligned}$$

Therefore, the conditional posterior distribution of β is given by,

$$\beta \mid \omega, \mathbf{y} \sim \mathcal{N}_p(\mathbf{b}^*, B^*) \quad (3.5.7)$$

where $\mathbf{b}^* = (X^\top \Omega X + B^{-1})^{-1} (X^\top \Omega \mathbf{z} + B^{-1} \mathbf{b})$ and $B^* = (X^\top \Omega X + B^{-1})^{-1}$.

The preservation of conjugacy under the Pólya-Gamma data augmentation provides the foundation for efficient posterior computation.

3.6 MCMC methods and convergence diagnostics

Markov chain Monte Carlo (MCMC) methods are employed throughout this dissertation for posterior computation. This section briefly outlines two most important methods.

3.6.1 Metropolis-Hastings Algorithm

The Metropolis-Hastings (MH) algorithm is a general MCMC method for generating samples from a target distribution $\pi(\cdot)$ with the help of some proposal distribution $q(\cdot \mid \cdot)$. The goal is to obtain samples $Y_1, \dots, Y_n \sim \pi$.

The algorithm is explained here.

Algorithm 1: Metropolis-Hastings Algorithm

Input: Target density $f(y)$, proposal density $q(\cdot \mid y)$, number of iterations n

Output: Samples $\{y_i\}_{i=1}^n$

Initialization: Set y_0 such that $f(y_0) > 0$

1 **for** $i = 1$ **to** n **do**

 // Propose a new candidate

2 Generate $y^* \sim q(\cdot \mid y_{i-1})$

 // Compute acceptance probability

3 Compute

$$\alpha = \min \left\{ 1, \frac{f(y^*) q(y_{i-1} \mid y^*)}{f(y_{i-1}) q(y^* \mid y_{i-1})} \right\}.$$

 // Draw a uniform random number

4 Generate $u \sim U(0, 1)$

5 **if** $u \leq \alpha$ **then**

 // Accept the proposal

6 Set $y_i = y^*$

7 **else**

 // Reject the proposal

8 Set $y_i = y_{i-1}$

9 **end**

10 **end**

11 **return** $\{y_i\}_{i=1}^n$

One of the major characteristics of MH algorithm is that we don't need to know the normalizing constant for the target density, which is very useful especially in bayesian context.

3.6.2 Gibbs sampler

Gibbs sampling helps us to draw samples from a *multivariate* target distribution. Suppose we want to draw n random samples $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_n$ from the joint target density $f(\cdot)$, where each $\mathbf{y}_i$ is a k -component vector. Let the t -th iteration of the i -th component be denoted by $y_i^{(t)}$, so that the state at iteration t is $\mathbf{y}^{(t)} = (y_1^{(t)}, y_2^{(t)}, \dots, y_k^{(t)})$. Conditional distribution for the component y_i given $y_1, y_2, \dots, y_{i-1}, y_{i+1}, \dots, y_n$ is denoted by $f_i(\cdot \mid y_1, y_2, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$.

Algorithm 2: Gibbs Sampler

Input: full conditional distributions $f_i(y_i \mid y_1, y_2, \dots, y_{i-1}, y_{i+1}, \dots, y_n)$ for $i = 1, \dots, k$, Joint target distribution $f(\mathbf{y})$, number of iterations n

Output: Samples $\{\mathbf{y}^{(t)}\}_{t=1}^n$

Initialization: Initial value $\mathbf{y}^{(0)}$ such that $f(\mathbf{y}^{(0)}) > 0$

```

1 for  $t = 1$  to  $n$  do
    // Sample each component from its full conditional
2   for  $i = 1$  to  $k$  do
3     Generate
        
$$y_i^{(t)} \sim f(y_i \mid (y_1^{(t)}, \dots, y_{i-1}^{(t)}, y_{i+1}^{(t-1)}, \dots, y_k^{(t-1)}))$$

        // Use most recent values: components 1 to  $i-1$  from current iteration,
        //  $i+1$  to  $k$  from previous
4   end
    // Store the complete vector
5   Set  $\mathbf{y}^{(t)} = (y_1^{(t)}, y_2^{(t)}, \dots, y_k^{(t)})$ 
6 end
7 return  $\{\mathbf{y}^{(t)}\}_{t=1}^n$ 

```

3.6.3 Convergence diagnostic

We can use various methods to assess whether or not stationarity has reached or not. Few methods are explained here.

- **Trace plot** : A trace plot is a plot of the iteration number against the value of the draw of the parameter at each iteration. Observing this, we can see whether the chain gets stuck in certain areas of the parameter space or not.
- **Autocorrelation plot** : Another way to assess convergence is to assess the autocorrelations between the draws of our Markov chain. We would expect k th lag autocorrelation to be smaller as k increases. High autocorrelation value for higher values of k indicates slow mixing.
- **Gelman-Rubin Diagnostic**: The Gelman-Rubin diagnostic $\hat{R}$ compares between-chain variance B to within-chain variance W to assess MCMC convergence. For m chains of length $2n$ with first n draws discarded, let θ_{ij} be the i th draw in the j th chain. Define $W = \frac{1}{m} \sum_{j=1}^m s_j^2$ and $B = \frac{n}{m-1} \sum_{j=1}^m (\bar{\theta}_j - \bar{\bar{\theta}})^2$. The estimated variance is $\widehat{Var}(\theta) = (1 - \frac{1}{n})W + \frac{1}{n}B$ and the **potential scale reduction factor** is $\hat{R} = \sqrt{\frac{\widehat{Var}(\theta)}{W}}$ and Values $\hat{R} < 1.1$ indicate convergence, otherwise we need to run the chains longer to improve convergence.

4 Prior specification

Motivated by the preceding discussion, we adopt the Polya-Gamma data augmentation strategy. Rather than working directly with the intractable logistic likelihood, we augment each y_i with an independent latent variable $\omega_i \sim \text{PG}(1, 0)$. Under this augmentation, the joint likelihood becomes Gaussian, greatly simplifying posterior computation.

This section explains the hierarchical shrinkage priors placed on the regression coefficients. We consider three classes of shrinkage priors : BGHS, GRASP and GIGG, and each of them entails some interesting characteristics and imposes a different form of regularization on the regression coefficient.

4.1 Bayesian Grouped Horseshoe (BGHS)

4.1.1 Motivation

The horseshoe prior [6] is a powerful tool for sparse estimation because it combines two desirable properties — an infinite spike at zero and heavy, Cauchy like tails. However the standard horseshoe treats all predictors *exchangeably*; it does not exploit any grouping structure. The BGHS extends the standard horseshoe to handle such grouped predictors by introducing a group-level shrinkage parameter alongside the usual local and global parameters.

4.1.2 Model structure

$$\begin{aligned}
 \mathbf{y} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \frac{(e^{\mathbf{x}_i^\top \boldsymbol{\beta}})^{y_i}}{1 + e^{\mathbf{x}_i^\top \boldsymbol{\beta}}} \\
 \beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
 \lambda_{gj} &\sim \mathcal{C}^+(0, 1) \\
 \delta_g &\sim \mathcal{C}^+(0, 1) \\
 \tau &\sim \mathcal{C}^+(0, 1)
 \end{aligned} \tag{4.1.1}$$

By section 3.5.3, model (4.1.1) can be equivalently written as the following :

$$\begin{aligned}
 \mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} - \frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
 \omega_i &\sim \text{PG}(1, 0), \kappa_i = y_i - 0.5 \\
 \beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
 \lambda_{gj} &\sim \mathcal{C}^+(0, 1) \\
 \delta_g &\sim \mathcal{C}^+(0, 1) \\
 \tau &\sim \mathcal{C}^+(0, 1)
 \end{aligned} \tag{4.1.2}$$

4.1.3 Full conditional posterior distributions

Using the hierarchy in (4.1.2), we develop a Gibbs sampler by deriving the full conditional posterior distribution. The derivation exploits the scale mixture representation of the half-Cauchy distribution (See Appendix C). Defining the auxiliary variables t_{gj} , c_g and ν , the model (4.1.2) can be written in the following form :

$$\begin{aligned}
\mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} \cdot - \frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
\omega_i &\sim PG(1, 0), \kappa_i = y_i - 0.5 \\
\beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
\lambda_{gj}^2 \mid t_{gj} &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{t_{gj}} \right) \quad t_{gj} \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
\delta_g^2 \mid c_g &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{c_g} \right), c_g \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
\tau^2 \mid \nu &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{\nu} \right), \nu \sim \text{IG} \left(\frac{1}{2}, 1 \right)
\end{aligned} \tag{4.1.3}$$

The posterior distribution is presented below; its detailed derivation is deferred to Appendix B.1.

$$\begin{aligned}
\omega_i \mid \cdot &\sim PG(1, \mathbf{x}_i^\top \boldsymbol{\beta}) \\
\boldsymbol{\beta} \mid \cdot &\sim \mathcal{N}_p \left((X^\top \Omega X + D^{-1})^{-1} X^\top \boldsymbol{\kappa}, (X^\top \Omega X + D^{-1})^{-1} \right) \\
\lambda_{gj}^2 \mid \cdot &\sim \text{IG} \left(1, \frac{\beta_{gj}^2}{2\tau^2 \delta_g^2} + \frac{1}{t_{gj}} \right) \\
t_{gj} \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{\lambda_{gj}^2} \right) \\
\delta_g^2 \mid \cdot &\sim \text{IG} \left(\frac{p_g + 1}{2}, \frac{1}{2\tau^2} \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{\lambda_{gj}^2} + \frac{1}{c_g} \right) \\
c_g \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{\delta_g^2} \right) \\
\tau^2 \mid \cdot &\sim \text{IG} \left(\frac{p+1}{2}, \frac{1}{2} \sum_{g=1}^G \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{\lambda_{gj}^2 \delta_g^2} + \frac{1}{\nu} \right) \\
\nu \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{\tau^2} \right)
\end{aligned} \tag{4.1.4}$$

where $D^{p \times p} = \text{diag}((\tau^2 \lambda_{gj}^2 \delta_g^2)), \Omega = \text{diag}(\omega_1, \omega_2, \dots, \omega_n)$ and $\boldsymbol{\kappa} = (\kappa_1, \kappa_2, \dots, \kappa_n)^\top$, $\kappa_i = y_i - 0.5$

4.2 Grouped Regression with Adaptive Shrinkage Prior (GRASP)

4.2.1 Motivation

The GRASP framework is a very recent work in the continuous bayesian shrinkage prior by Tew et al. [19]. It assigns a *normal beta prime* prior to both local and group shrinkage parameters. The NBP prior is a flexible, adaptive generalization of horseshoe prior with tunable hyperparameters that control the tail behavior, enabling a flexible range of sparsity – from strong shrinkage to ridge-like shrinkage.

A major contribution of GRASP is a novel framework to explicitly quantify the apriori correlation between shrinkage parameters within a group (Proposition 3.1 of [19]). This apriori correlations are controlled by the hyperparameters, we can choose some value of them, or can take appropriate hyperprior on them. GRASP can produce different prior correlation distributions, ranging from highly individualistic (correlation near 0) to highly group-wise (correlation near 1). Here, fully bayesian setup is considered where we put a half-Cauchy priors on the hyperparameters.

4.2.2 Model structure

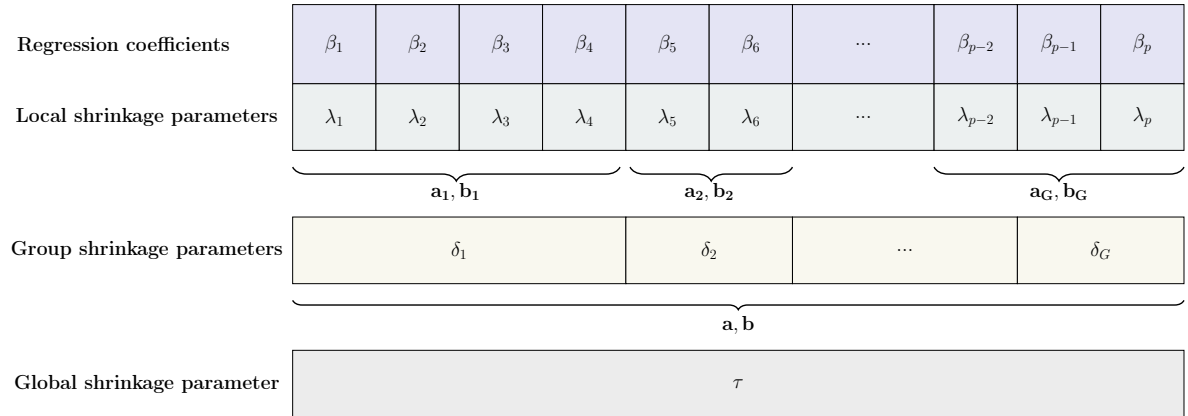


Figure 2: Hierarchical shrinkage framework for the GRASP model. The braces indicate the grouping structure of the local shrinkage parameters λ and the group shrinkage parameters δ , with their respective hyperparameters (a_g, b_g) and (a, b) .

$$\begin{aligned}
 \mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} \cdot -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
 \omega_i &\sim PG(1, 0), \kappa_i = y_i - 0.5 \\
 \beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
 \lambda_{gj}^2 &\sim \mathcal{B}'(a_g, b_g) \\
 \delta_g^2 &\sim \mathcal{B}'(a, b) \\
 \tau &\sim \mathcal{C}^+(0, 1) \\
 a_g &\sim \mathcal{C}^+(0, 1), b_g \sim \mathcal{C}^+(0, 1) \\
 a &\sim \mathcal{C}^+(0, 1), b \sim \mathcal{C}^+(0, 1)
 \end{aligned} \tag{4.2.1}$$

4.2.3 Full conditional posterior distributions

Scale mixture representation in Appendix C lead us to another representation of (4.2.1) as follows :

$$\begin{aligned}
\mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} \cdot -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
\omega_i &\sim PG(1, 0), \kappa_i = y_i - 0.5 \\
\beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
\lambda_{gj}^2 \mid t_{gj} &\sim \text{IG} \left(b_g, \frac{1}{t_{gj}} \right), t_{gj} \sim \text{IG}(a_g, 1) \\
\delta_g^2 \mid c_g &\sim \text{IG} \left(b, \frac{1}{c_g} \right), c_g \sim \text{IG}(a, 1) \\
\tau^2 \mid \nu &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{\nu} \right), \nu \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
a_g^2 \mid d_g &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{d_g} \right), d_g \sim \text{IG} \left(\frac{1}{2}, 1 \right); b_g^2 \mid e_g \sim \text{IG} \left(\frac{1}{2}, \frac{1}{e_g} \right), e_g \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
a^2 \mid d &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{d} \right), d \sim \text{IG} \left(\frac{1}{2}, 1 \right); b^2 \mid e \sim \text{IG} \left(\frac{1}{2}, \frac{1}{e} \right), e \sim \text{IG} \left(\frac{1}{2}, 1 \right) \quad (4.2.2)
\end{aligned}$$

The full conditional distributions are given as :

$$\begin{aligned}
\omega_i \mid \cdot &\sim PG(1, \mathbf{x}_i^\top \boldsymbol{\beta}) \\
\boldsymbol{\beta} \mid \cdot &\sim \mathcal{N}_p \left((X^\top \Omega X + D^{-1})^{-1} X^\top \boldsymbol{\kappa}, (X^\top \Omega X + D^{-1})^{-1} \right) \\
\lambda_{gj}^2 \mid \cdot &\sim \text{IG} \left(b_g + \frac{1}{2}, \frac{1}{t_{gj}} + \frac{\beta_{gj}^2}{2\tau^2 \delta_g^2} \right) \\
t_{gj} \mid \cdot &\sim \text{IG} \left(a_g + b_g, 1 + \frac{1}{\lambda_{gj}^2} \right) \\
\delta_g^2 \mid \cdot &\sim \text{IG} \left(b + \frac{p_g}{2}, \frac{1}{c_g} + \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2 \lambda_{ij}^2} \right) \\
c_g \mid \cdot &\sim \text{IG} \left(a + b, 1 + \frac{1}{\delta_g^2} \right) \\
\tau^2 \mid \cdot &\sim \text{IG} \left(\frac{p+1}{2}, \frac{1}{\nu} + \frac{1}{2} \sum_{g=1}^G \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{\lambda_{gj}^2 \delta_g^2} \right) \\
\nu \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{\tau^2} \right) \\
d_g \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{a_g^2} \right); e_g \mid \cdot \sim \text{IG} \left(1, 1 + \frac{1}{b_g^2} \right) \\
d \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{a^2} \right); e \mid \cdot \sim \text{IG} \left(1, 1 + \frac{1}{b^2} \right)
\end{aligned}$$

$$\begin{aligned}
a_g^2 \mid \cdot &\sim \left(\frac{1}{\Gamma(a_g)} \right)^{p_g} \left\{ \prod_{j=1}^{p_g} t_{gj} \right\}^{-a_g-1} (a_g^2)^{-3/2} \exp \left\{ -\frac{1}{d_g a_g^2} \right\} \\
b_g^2 \mid \cdot &\sim \prod_{j=1}^{p_g} \left\{ \left(\frac{1}{t_{gj}} \right)^{b_g} \frac{1}{\Gamma(b_g)} (\lambda_{gj}^2)^{-b_g-1} \right\} (b_g^2)^{-3/2} \exp \left\{ -\frac{1}{e_g b_g^2} \right\} \\
a^2 \mid \cdot &\sim \left(\frac{1}{\Gamma(a)} \right)^G \left(\prod_{g=1}^G c_g \right)^{-a-1} (a^2)^{-3/2} \exp \left\{ -\frac{1}{d a^2} \right\} \\
b^2 \mid \cdot &\sim \left(\frac{1}{\Gamma(b)} \right)^G \left(\prod_{g=1}^G \delta_g^2 \right)^{-b-1} \left(\prod_{g=1}^G \frac{1}{c_g} \right)^b (b^2)^{-3/2} \exp \left\{ -\frac{1}{e b^2} \right\} \quad (4.2.3)
\end{aligned}$$

where $D^{p \times p} = \text{diag}((\tau^2 \lambda_{gj}^2, \delta_g^2)), \Omega = \text{diag}(\omega_1, \omega_2, \dots, \omega_n)$ and $\boldsymbol{\kappa} = (\kappa_1, \kappa_2, \dots, \kappa_n)^\top$, $\kappa_i = y_i - 0.5$

The full derivation is deferred to Appendix B.2.

4.3 Group Inverse-Gamma Gamma Prior (GIGG)

4.3.1 Motivation

The GIGG prior was developed by Boss et al. (2016) [5] to overcome the limitation of the group horseshoe prior which lacks a hyperparameter to control how *dependent the shrinkage is within a group*. GIGG introduces explicit, data-adaptive control over the trade-off between group level and individual level shrinkage. It introduces two key hyperparameters a_g and b_g for group and local shrinkage parameter, such that the induced prior on the product of the group and local shrinkage parameters yields the desired marginal shrinkage profile. These hyperparameters are illustrated in Figure 3.

4.3.2 Model structure

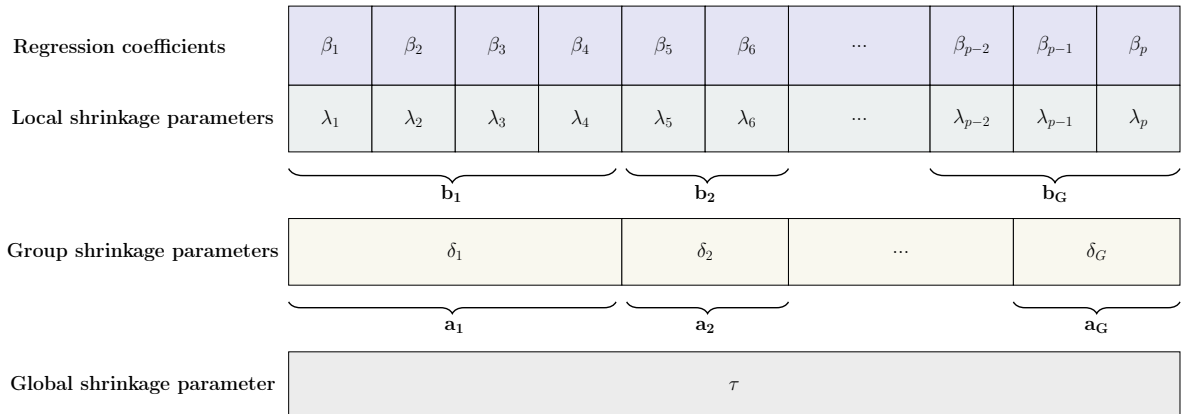


Figure 3: Hierarchical shrinkage framework for the GIGG model. The braces indicate the grouping structure of the local shrinkage parameters λ and the group shrinkage parameters δ , with their respective hyperparameters b_g and a_g .

$$\begin{aligned}
\mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} \cdot -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
\omega_i &\sim PG(1, 0), \kappa_i = y_i - 0.5 \\
\beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
\lambda_{gj}^2 &\sim \text{IG}(b_g, 1) \\
\delta_g^2 &\sim \text{G}(a_g, 1) \\
\tau &\sim \mathcal{C}^+(0, 1) \\
a_g &\sim \mathcal{C}^+(0, 1) \\
b_g &\sim \mathcal{C}^+(0, 1)
\end{aligned} \tag{4.3.1}$$

4.3.3 Full conditional posterior distributions

Using the scale mixture representation of half Cauchy distribution, model from (4.3.1) can be written as following :

$$\begin{aligned}
\mathbf{y}, \boldsymbol{\omega} \mid X, \boldsymbol{\beta} &\sim \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} \cdot -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
\omega_i &\sim PG(1, 0), \kappa_i = y_i - 0.5 \\
\beta_{gj} \mid \tau^2, \lambda_{gj}, \delta_g &\sim \mathcal{N}(0, \tau^2 \lambda_{gj}^2 \delta_g^2) \\
\lambda_{gj}^2 &\sim \text{IG}(b_g, 1) \\
\delta_g^2 &\sim \text{G}(a_g, 1) \\
\tau^2 \mid \nu &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{\nu} \right) ; \nu \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
a_g^2 \mid d_g &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{d_g} \right) ; d_g \sim \text{IG} \left(\frac{1}{2}, 1 \right) \\
b_g^2 \mid e_g &\sim \text{IG} \left(\frac{1}{2}, \frac{1}{e_g} \right) ; e_g \sim \text{IG} \left(\frac{1}{2}, 1 \right)
\end{aligned} \tag{4.3.2}$$

Using (4.3.2), full conditional posterior distributions are given by :

$$\begin{aligned}
\omega_i \mid \cdot &\sim PG(1, \mathbf{x}_i^\top \boldsymbol{\beta}) \\
\boldsymbol{\beta} \mid \cdot &\sim \mathcal{N}_p \left((X^\top \Omega X + D^{-1})^{-1} X^\top \boldsymbol{\kappa}, (X^\top \Omega X + D^{-1})^{-1} \right) \\
\lambda_{gj}^2 \mid \cdot &\sim \text{IG} \left(b_g + \frac{1}{2}, 1 + \frac{\beta_{gj}^2}{2\tau^2 \delta_g^2} \right) \\
\delta_g^2 \mid \cdot &\sim \text{GIG} \left(a_g - \frac{p_g}{2}, 2, \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{\lambda_{gj}^2 \tau^2} \right)
\end{aligned}$$

$$\begin{aligned}
\tau^2 \mid \cdot &\sim \text{IG} \left(\frac{p+1}{2}, \frac{1}{\nu} + \frac{1}{2} \sum_{g=1}^G \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{\lambda_{gj}^2 \delta_g^2} \right) \\
\nu \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{\tau^2} \right) \\
d_g \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{a_g^2} \right) \\
e_g \mid \cdot &\sim \text{IG} \left(1, 1 + \frac{1}{b_g^2} \right) \\
a_g^2 \mid \cdot &\sim \frac{(\delta_g^2)^{a_g-1}}{\Gamma(a_g)} (a_g^2)^{-3/2} \exp \left\{ -\frac{1}{d_g a_g^2} \right\} \\
b_g^2 \mid \cdot &\sim \left(\frac{1}{\Gamma(b_g)} \right)^{p_g} \left\{ \prod_{j=1}^{p_g} \lambda_{gj}^2 \right\}^{-b_g-1} (b_g^2)^{-3/2} \exp \left\{ -\frac{1}{e_g b_g^2} \right\}
\end{aligned} \tag{4.3.3}$$

where $D^{p \times p} = \text{diag}((\tau^2 \lambda_{gj}^2 \delta_g^2)), \Omega = \text{diag}(\omega_1, \omega_2, \dots, \omega_n)$ and $\kappa = (\kappa_1, \kappa_2, \dots, \kappa_n)^\top$, $\kappa_i = y_i - 0.5$

Full derivation is given in the Appendix B.3.

4.4 Comparison of GRASP and GIGG

A key distinction between the GIGG and GRASP frameworks lies in the treatment of the group shrinkage parameter and the resulting implications for within-group dependence.

Note that, for GIGG model (4.3.1) implies that induced prior on the product is a beta prime prior i.e. $\delta_g^2 \lambda_{gj}^2 \sim \mathcal{B}'(a_g, b_g)$. The choice of the inverse-gamma prior for the local shrinkage parameters λ_{gj}^2 is driven by its *heavy tailed nature* as compared to the gamma distribution. This heavy tail allows individual coefficients to “escape” i.e. it prevents the over regularization of the large, non-null coefficients. The gamma prior on group shrinkage parameters acts as a “sharing” mechanism and it introduces a within-group dependence. As the prior variance of β_{gj} is :

$$\text{Var}(\beta_{gj}) = \tau^2 \lambda_{gj}^2 \delta_g^2$$

Since, δ_g^2 is common to all coefficients in group g . So, if δ_g^2 is large, all coefficients within that group have large prior variances and are all allowed to be large, hence the group is *active* as a whole. If δ_g^2 is small, the exact reverse logic establishes the group will be *inactive* as a whole.

From the GRASP and GIGG models in (4.2.1) and (4.3.1) respectively, one can observe that ideological difference in the group shrinkage parameters. Former one defines that the group shrinkage parameters has global hyperparameters (a and b), but latter one defines that the distribution of group shrinkage parameters is independent but not identical, i.e. it depends on that specific group.

GRASP assumes that the *overall nature of group-level shrinkage* is a global property, not something that varies wildly across groups. GIGG assumes that *different groups may have fundamentally different shrinkage requirements*.

5 Experimental settings

We performed a comparative evaluation of BGHS, GRASP and GIGG for logistic regression based on few controlled conditions. All experiments were conducted in using the **R** statistical software. In this section, we describe the design of the simulation study.

5.1 Overview

The simulation study is structured around four experimental scenarios that represent different types of signal distribution within groups. For each scenario, we vary the within-group and between-group correlation structures. The true regression coefficients are fixed for each scenario, the response variable is generated from a logistic regression model. Each combination of experimental setting and correlation structure is replicated multiple number of times to obtain stable estimates of the evaluation metrics.

5.2 Data generating process

For each replication, we generate a design matrix $X \in \mathbb{R}^{n \times p}$ by drawing n samples from p -variate normal distribution with mean vector $\mathbf{0}$ and correlation matrix Σ . The correlation matrix is constructed to reflect the group structure.

5.2.1 Correlation structure

The predictors are partitioned into G groups. The correlation between two predictors is defined as :

$$\text{cor}(\mathbf{x}_j, \mathbf{x}_{j'}) = \begin{cases} \rho_w & \text{if } g = g', j \neq j' \\ \rho_b & \text{if } g \neq g' \\ 1 & \text{if } g = g', j = j' \end{cases} \quad (5.2.1)$$

where, $\mathbf{x}_j$ and $\mathbf{x}_{j'}$ denotes the j and j' th column of design matrix X respectively. Note that, former one belongs to the g th group and latter one belongs to g' th.

Three correlation pairs are considered :

$$(\rho_w, \rho_b) \in \{(0.8, 0.2), (0.5, 0.1), (0.3, 0.01)\}$$

These value ranges from high within-group correlation with moderate between-group correlation to low within-group correlation with negligible between-group correlation.

5.2.2 True coefficient vectors

For each setting, the true coefficient vector β is specified through a predetermined sparsity pattern and signal configuration. Section 5.3 comprises all the details.

5.2.3 Response generation

Given the design matrix X and true coefficient vector β , the linear predictor is computed as :

$$\eta = X\beta = (\mathbf{x}_1^\top \beta, \mathbf{x}_2^\top \beta, \dots, \mathbf{x}_n^\top \beta)^\top$$

The probability of success for each observation is then :

$$\pi_i = \frac{\exp\{\mathbf{x}_i^\top \boldsymbol{\beta}\}}{1 + \exp\{\mathbf{x}_i^\top \boldsymbol{\beta}\}}, i = 1, 2, \dots, n \quad (5.2.2)$$

Finally, the binary response y_i is generated independently from $Bernoulli(\pi_i)$. The data generation procedure is concisely summarized in Algorithm 3.

Algorithm 3: Data Generation for Simulation Study

Input: Number of observations n , number of predictors p , number of groups G , group sizes $\{p_g\}_{g=1}^G$, within-group correlation ρ_w , between-group correlation ρ_b , true coefficient vector $\boldsymbol{\beta}$

Output: Design matrix $X \in \mathbb{R}^{n \times p}$, binary response vector $\mathbf{y} \in \{0, 1\}^n$

// Step 1: Construct the correlation matrix

1 Construct $\Sigma \in \mathbb{R}^{p \times p}$ using (5.2.1) // Step 2: Generate the design matrix

2 Generate $X \in \mathbb{R}^{n \times p}$ by drawing n independent samples from

$$X \sim \mathcal{N}_p(\mathbf{0}, \Sigma).$$

Center and scale each column of X to have mean zero and unit variance.

// Step 3: Compute the linear predictor

3 Compute the linear predictor

$$\boldsymbol{\eta} = X\boldsymbol{\beta} = (\mathbf{x}_1^\top \boldsymbol{\beta}, \mathbf{x}_2^\top \boldsymbol{\beta}, \dots, \mathbf{x}_n^\top \boldsymbol{\beta})^\top$$

4 for $i = 1$ to n do

 // Step 4: Compute success probabilities

5 Calculate π_i using (5.2.2)

 // Step 5: Generate binary responses

6 Generate

$$y_i \sim \text{Bernoulli}(\pi_i).$$

7 end

8 return $X, \mathbf{y}$

5.3 Experimental settings

In this section, the experimental settings will be discussed under which we have evaluated the performance of the three grouped shrinkage priors. Each setting is given a unique name reflecting its sparsity structure, as outlined below. Complete details of each experimental settings are concisely summarized in Table 1.

1. **Concentrated** : Exactly one predictor per group is active. There are five groups, each of size 10, giving $p = 50$. The active coefficients are 0.5, 1, 1.5, 2, 2 across the five groups, with all other coefficients set to zero.
2. **Distributed** : All predictors in the first group are active, with coefficient 0.5 for each of the 10 predictors in that group. It consists of five groups, and remaining four groups are entirely null.

3. **Dense** : This involves heavily dense signals in all six groups. The group sizes are 30, 10, 20, 5, 15 and 20. Also, the active coefficients vary in magnitude and are randomly sampled from specified sets.
4. **Half** : This experimental setting features a combination of dense signals in half of the groups and few to no signal regressors in the other half. It consists of seven groups of varying sizes, with some groups being dense, some sparse, and some entirely null. The group sizes are 25, 10, 10, 10, 5, 15 and 5, with a total of $p = 80$ predictors. The scenario represents a heterogeneous setting where groups exhibit different sparsity patterns.

Group	p = 50, n = 500				p = 100, n = 300		p = 80, n = 200	
	Concentrated		Distributed		Dense		Half	
	AV/GS	Coef	AV/GS	Coef	AV/GS	Coef	AV/GS	Coef
1	1/10	0.5	10/10	0.5	27/30	[3, 5, 8]	22/25	0.8
2	1/10	1	0/10	–	8/10	[2, 3, 4, 5]	0/10	–
3	1/10	1.5	0/10	–	18/20	7.5	3/10	2.5
4	1/10	2	0/10	–	4/5	[9, 5, 8, 7]	8/10	1.5
5	1/10	2	0/10	–	14/15	1.5	0/5	–
6	–	–	–	–	18/20	0.5	4/15	[1, 2, 3, 5]
7	–	–	–	–	–	–	1/5	2

Table 1: Experimental settings considered in the simulation study

Note: AV and GS denote the number of active variables and the group size, respectively.

6 Methodology

This section describes the computational and evaluation procedures used to compare the three grouped shrinkage priors – BGHS, GRASP and GIGG. For each experimental setting described in the previous section, we apply these three methods to the generated datasets and evaluate their performance using two primary metrics: **estimation accuracy** (via mean square error of coefficients) and **variable selection performance** (via ROC-AUC). The following subsections detail the MCMC implementation, convergence diagnostics and evaluation metrics.

6.1 MCMC implementation

For each of the three hierarchical models, we have already discussed the full conditional posterior distributions in (4.1.4), (4.2.3) and (4.3.3).

Three separate functions are created, each designated to collect posterior samples for one of the three hierarchies. The corresponding functions are provided in Appendix E. These functions are iteratively sample from the full conditional distributions, which are all available in closed form for BGHS. But for GRASP and GIGG, however certain hyperparameters do not have standard conditional distributions from which direct sampling is possible. Specifically : a_g^2, b_g^2, a^2, b^2 for GRASP and a_g^2, b_g^2 for GIGG. (See last four distributions in (4.2.3) and (4.3.3)).

For these parameters, we employ a **Metropolis-within-Gibbs** (MHwG) procedure.

6.1.1 MHwG procedure

This involves embedding a MH step inside the Gibbs sampler to update each those parameters individually, conditioned on all other parameters. For each of the Gibbs iteration, we have drawn 1001 samples from those particular distributions in MH step and considers 1001th sample as a sample from that distribution.

Here, in the MHwG process for the corresponding parameters **lognormal** distribution is considered as the proposal.

As log-normal distribution is taken as proposal for every parameters, so even if the parameter has different distributions the overall idea will be same. To illustrate this, consider the update of a_g^2 in the GRASP model. The full conditional posterior of a_g^2 from (4.2.3) is given by :

$$p(a_g^2 | \cdot) \propto \left(\frac{1}{\Gamma(a_g)} \right)^{p_g} \left\{ \prod_{j=1}^{p_g} t_{gj} \right\}^{-a_g-1} (a_g^2)^{-3/2} \exp \left\{ -\frac{1}{d_g a_g^2} \right\}$$

Define, the sample at $(i-1)$ th iteration is a_{gi-1}^2 and draw the proposal value a_g^{*2} from $LN(a_{gi-1}^2, \sigma^2) := q(\cdot | a_{gi-1}^2)$.

Algorithm 4: MH step in MHwG for a_g^2 is GRASP

Input: Target density $p(a_g^2 | \cdot)$, proposal density $LN(a_g^2, \sigma^2)$, number of iterations 1001

Output: Samples $\{a_{gi}^2\}_{i=1}^{1001}$

Initialization: Set a_{g0}^2 such that $p(a_{g0}^2 | \cdot) > 0$

1 **for** $i = 1$ **to** 1001 **do**

 // Propose a new candidate

2 Generate $a_g^{*2} \sim LN(a_{gi-1}^2, \sigma^2)$

 // Compute $\log \alpha$ from (6.1.1)

3 Compute

$$\log \alpha = \min \left\{ 0, \log \left(\frac{p(a_g^{*2} | \cdot) a_g^{*2}}{p(a_{gi-1}^2 | \cdot) a_{gi-1}^2} \right) \right\} = \min\{0, \log \alpha^*\}$$

 // Draw a uniform random number

4 Generate $u \sim U(0, 1)$

5 **if** $\log u \leq \log \alpha$ **then**

 // Accept the proposal

6 Set $a_{gi}^2 = a_g^{*2}$

7 **else**

 // Reject the proposal

8 Set $a_{gi}^2 = a_{gi-1}^2$

9 **end**

10 **end**

11 **return** $\{a_{gi}^2\}_{i=1}^{1001}$

The acceptance probability (ignoring the min and 1 for now) :

$$\begin{aligned}
\alpha^* &= \frac{p(a_g^{*2} \mid \cdot) a_g^{*2}}{p(a_{gi-1}^2 \mid \cdot) a_{gi-1}^2} \\
&= \left(\frac{\Gamma(a_g^*)}{\Gamma(a_{gi-1})} \right)^{-p_g} \left(\prod_{j=1}^{p_g} t_{gj} \right)^{-(a_g^* - a_{gi-1})} \left(\frac{a_g^{*2}}{a_{gi-1}^2} \right)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{d_g} \left(\frac{1}{a_g^{*2}} - \frac{1}{a_{gi-1}^2} \right) \right\} \frac{a_g^{*2}}{a_{gi-1}^2} \\
\Rightarrow \log(\alpha^*) &= -p_g \left\{ \log \Gamma(a_g^*) - \log \Gamma(a_{gi-1}) \right\} - (a_g^* - a_{gi-1}) \sum_{j=1}^{p_g} \log(t_{gj}) \\
&\quad - \frac{3}{2} (\log a_g^{*2} - \log a_{gi-1}^2) - \frac{1}{d_g} \left(\frac{1}{a_g^{*2}} - \frac{1}{a_{gi-1}^2} \right) + (\log a_g^{*2} - \log a_{gi-1}^2)
\end{aligned} \tag{6.1.1}$$

log of actual acceptance probability is given by :

$$\log \alpha = \min\{0, \log \alpha^*\}$$

Similarly, algorithm 4 can be easily tweaked to draw samples from the other distributions for which MHwG is required.

6.1.2 Tuning the σ^2 in proposal density

The efficiency of the MHwG step depends on the choice of σ^2 .

The choice of σ is very necessary as if σ^2 is large then the proposal variability will be larger. The acceptance probability reduces and hence the *chain slowly moves*. If it is small then proposed values are very close to the current value, leading to a high acceptance rate, which ultimately causes *high autocorrelation* in the chain. The whole procedure is summarized below :

- To select the appropriate σ , we perform a grid-search method. For each parameter, we consider a range of candidate σ values i.e. σ in 0.05 to 5.
- For each candidate σ , we run a short pilot MCMC chain to record the acceptance rate for the MH step of the parameter being tuned.
- We choose the σ that gives the acceptance rate closest to the target of 0.3 to 0.4. If the acceptance rate is too high, we increase σ ; if it is too low, we decrease the σ .
- Previous steps are repeated multiple times and for a particular tuning we have observed multiple times that which σ is optimal for that grid. We have considered the σ that reveals as optimal in maximum of the replications.
- Once the optimal σ is identified, it is fixed and used in the final, long MCMC run for that experimental setting.

But, one thing we need to remember, by **Acceptance probability** we are not meaning the α notation defined previously. Acceptance probability in this tuning section means *the fraction of proposed moves which are accepted*. So, we are basically want a moderate acceptance probability here, that's why the threshold is 0.3 to 0.4.

Prior	$\sigma_{a_g^2}$	$\sigma_{b_g^2}$	σ_{a^2}	σ_{b^2}
GRASP	1.5	1.5	3	1.5
GIGG	5	1.5	—	—

Table 2: Tuned values of σ^2

6.1.3 Choice of burn-in period, total iterations and thinning

Before proceeding with the final posterior inference, we must determine the appropriate MCMC settings—namely, the burn-in period, the total number of iterations, and the thinning interval—for each of the three samplers under each experimental setting. These parameters are not fixed a priori; they depend on the mixing behaviour of the chain, which varies across methods, scenarios, and correlation structures. A systematic procedure is therefore employed to select these values in a data adaptive manner.

For each of the method and scenario, we first run a pilot chain with a large number of iterations (e.g. 50000) and no burn-in. We have used the tuned proposal parameter in MHwG steps. From this chain, we have observed $ESS > 400$ or not for each β_j using the `coda` package in R.

For choosing the burn-in period, we have visually inspect trace plots of a random subset of β_j 's and several other parameters. Before that, we have drawn two chains from overdispersed initial values and observe the psrf $\hat{R}$ is less than 1.1 or not for all β_j 's after discarding the burn-in. Based on this diagnostics, we find that a burn-in of 10000 – 20000 iterations is sufficient for most settings, with longer burn-in required for more complex experimental settings.

We also computed the ACF plots for a representative subset of coefficients to assess the rate at which it decays. From our inspection, it decays quickly so thinning is not required here. The final MCMC parameters for each method and scenario are summarised in table.

Scenario	BGHS	GRASP	GIGG
1	20k/10k	20k/10k	20k/10k
2	20k/10k	30k/20k	20k/10k
3	40k/20k	30k/20k	50k/20k
4	40k/20k	40k/20k	60k/30k

Table 3: Number of samples (in thousands) and burn-in iterations used for each experimental setting, written as **total/burn-in** (e.g. 20k/10k means 20000 total iterations with 10000 burn-in)

6.2 Evaluation metrics

To compare the estimation and variable selection performance of the three grouped shrinkage priors – BGHS, GRASP and GIGG – we employ two primary evaluation metrics : **MSE** stratified by null and non-null coefficients and **ROC-AUC** curves. These metrics are computed for each replication and then averaged across replications to obtain stable summary statistics.

6.2.1 Estimation accuracy : MSE

The first metric assesses the *estimation accuracy* of the regression coefficients. This method evaluates the MSE stratified by null and non-null coefficients. The following steps were carried out to evaluate this measure:

- Starting with a method and a correlation structure, the aforementioned MCMC parameters, tuning value of σ^2 are used to generate posterior samples of $\beta_j, j = 1, 2, \dots, p$. Note that, p can vary with respect to the experimental settings (See table 1).
- Posterior means of each β_j is considered as the estimate and it is denoted by $\hat{\beta}_j$. Define, $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_p)$, which denotes the posterior mean estimate of the regression coefficients and this $\hat{\beta}$ can be partitioned as

$$\hat{\beta} = (\hat{\beta}_{null}, \hat{\beta}_{nonnull})$$

corresponding to the estimates of β for which the actual values are 0 and non-zero respectively.

- Define, $S_0 = \{j : \beta_j = 0\}$ and $S_1 = \{j : \beta_j \neq 0\}$, i.e, S_0 and S_1 be the collection of indices for null and non-null coefficients. Also, let p_0 and p_1 be the number of null and non-null coefficients, that is, $p_0 = |S_0|$ and $p_1 = |S_1|$, $p_0 + p_1 = p$.
- For a single replication, the following expressions are calculated :

$$\begin{aligned} \text{MSE}_{null} &= \frac{1}{p_0} \sum_{j \in S_0} \hat{\beta}_j^2, \quad \text{MSE}_{nonnull} = \frac{1}{p_1} \sum_{j \in S_1} (\hat{\beta}_j - \beta_j)^2 \\ \text{MSE}_{total} &= \frac{p_0}{p_0 + p_1} \text{MSE}_{null} + \frac{p_1}{p_0 + p_1} \text{MSE}_{nonnull} \end{aligned}$$

- This whole procedure is replicated for R times. Using the last steps, for r th replication, we will obtain : $\text{MSE}_{null}^{(r)}, \text{MSE}_{nonnull}^{(r)}$ and $\text{MSE}_{total}^{(r)}, r = 1, 2, \dots, R$.
- For each experimental setting - correlation structure - grouped hierarchy method triplet, we will get :

$$\begin{aligned} \overline{\text{MSE}}_{null} &= \frac{1}{R} \sum_{r=1}^R \text{MSE}_{null}^{(r)} \\ \overline{\text{MSE}}_{nonnull} &= \frac{1}{R} \sum_{r=1}^R \text{MSE}_{nonnull}^{(r)} \\ \overline{\text{MSE}}_{total} &= \frac{1}{R} \sum_{r=1}^R \text{MSE}_{total}^{(r)} \end{aligned}$$

6.2.2 Predictive performance : ROC curves

To evaluate the predictive performance of the three grouped shrinkage priors, we use ROC curves and area under the ROC curve (AUC). These are standard tools for assessing the discriminative ability of a binary classifier.

The ROC curve is constructed by varying the classification threshold on the predicted probabilities. For each threshold, we compute the sensitivity (true positive rate) and specificity (true negative rate), and plot sensitivity against (1-specificity).

- For each replication, we randomly split the data into a training set (70% of the observations) and a test set (the remaining 20%). The test set is held out and used only for evaluation.
- Before making predictions, we apply a *variable selection* step based on credible intervals. For each coefficient β_j , we compute its 95% credible interval from the posterior samples obtained on the training data.

A coefficient is classified as *active* if its credible interval does not contain 0 hence,

$$Active_j = \mathbf{1} \{CI_{95\%}(\beta_j) \text{ does not contain } 0\}$$

For the selected active set $\mathcal{S} = \{j : Active_j = 1\}$, we retain the corresponding posterior means $\hat{\beta}_j$. All non-selected candidates are set to zero. This yields a sparse posterior mean vector $\hat{\hat{\beta}}_j$ defined as :

$$\hat{\hat{\beta}}_j = \begin{cases} \hat{\beta}_j & \text{if } j \in \mathcal{S} \\ 0 & \text{if } j \notin \mathcal{S} \end{cases}$$

- Using $\hat{\hat{\beta}} = (\hat{\hat{\beta}}_1, \hat{\hat{\beta}}_2, \dots, \hat{\hat{\beta}}_p)^\top$, we estimate the predicted probability as

$$\hat{\pi}_i = \frac{\exp\{\mathbf{x}_i^\top \hat{\hat{\beta}}\}}{1 + \exp\{\mathbf{x}_i^\top \hat{\hat{\beta}}\}}$$

For each test observation i , we also have the true binary outcome $y_i \in \{0, 1\}$.

- For a threshold $t \in [0, 1]$, we classify observation i as 1 if $\hat{\pi}_i > t$ and classify as 0 otherwise. The classification results are then compared to the true outcomes y_i to compute :

$$\text{Sensitivity}(t) = \frac{\#\{i : y_i = 1, \hat{\pi}_i > t\}}{\#\{i : y_i = 1\}}, \text{ Specificity}(t) = \frac{\#\{i : y_i = 0, \hat{\pi}_i \leq t\}}{\#\{i : y_i = 0\}}$$

i.e. sensitivity is the proportion of actual positives correctly identified and specificity is the proportion of actual negatives correctly identified.

- By varying the threshold t over a fine grid of values, we obtain a sequence of (1-specificity, sensitivity) pairs. The curve starts at (0,0) at $t = 1$ and ends at (1,1) when $t = 0$. The AUC is basically the area under the curve, here that integral is approximated using the trapezoidal rule.
- For each of the 100 replications, we obtain a full ROC curve and its corresponding AUC. To produce a stable summary : at each threshold, we average the sensitivity and specificity across all replications. This yields mean ROC curve. The AUC is averaged across replications, and the standard deviation is reported.

7 Simulation results

7.1 Stratified MSE

In this section, we present the results of the simulation study comparing three grouped shrinkage priors—BGHS, GRASP and GIGG—across four experimental settings and three correlation structures. The results are obtained following the steps explained in Section 6. The primary goal is to assess how each method performs across different signal patterns (concentrated, distributed, dense and mixed) and correlation structures. All the MSE values are represented in the following table. Table 4, 5 and 6 represents the MSE value for null coefficients, non-null coefficients and total MSE respectively.

The results are discussed in three parts : null MSE, non-null MSE and total MSE. For each setting, we interpret the performance of the methods, drawing connections to the theoretical properties of the priors.

(ρ_w, ρ_b)	Prior	MSE for Null Coefficients			
		Setting 1	Setting 2	Setting 3	Setting 4
(0.8, 0.2)	BGHS	0.008 (0.001)	0.1283 (0.002)	5.200 (0.520)	11.800 (0.280)
	GRASP	0.006 (0.001)	0.0089 (0.001)	2.800 (0.280)	4.200 (0.120)
	GIGG	0.008 (0.001)	0.0030 (0.0008)	5.600 (0.560)	4.500 (0.450)
(0.5, 0.1)	BGHS	0.004 (0.0004)	0.011 (0.0001)	8.000 (0.800)	25.091 (2.510)
	GRASP	0.005 (0.0005)	0.005 (0.00005)	4.000 (0.400)	3.340 (2.020)
	GIGG	0.006 (0.0006)	0.0015 (0.0002)	2.500 (0.250)	2.400 (0.240)
(0.3, 0.01)	BGHS	0.003 (0.0003)	0.0098 (0.0001)	3.200 (0.320)	6.500 (0.650)
	GRASP	0.004 (0.0004)	0.0012 (0.0001)	1.200 (0.120)	1.500 (0.150)
	GIGG	0.005 (0.0005)	0.0010 (0.0001)	0.800 (0.080)	2.000 (0.200)

Table 4: Empirical MSE for null coefficients under different simulation settings and correlation structures. Standard deviations across replications are shown in parentheses.

(ρ_w, ρ_b)	Prior	MSE for Non-null Coefficients			
		Setting 1	Setting 2	Setting 3	Setting 4
(0.8, 0.2)	BGHS	0.140 (0.021)	0.100 (0.015)	25.020 (3.750)	27.200 (4.080)
	GRASP	0.120 (0.018)	0.095 (0.014)	14.513 (2.175)	3.600 (0.540)
	GIGG	0.190 (0.029)	0.150 (0.023)	14.511 (2.190)	19.800 (2.970)
(0.5, 0.1)	BGHS	0.064 (0.010)	0.060 (0.009)	31.000 (4.650)	80.000 (12.000)
	GRASP	0.095 (0.014)	0.050 (0.008)	14.800 (2.220)	200.000 (30.000)
	GIGG	0.106 (0.016)	0.032 (0.005)	13.200 (1.980)	15.700 (2.355)
(0.3, 0.01)	BGHS	0.043 (0.006)	0.030 (0.005)	10.500 (1.575)	83.500 (12.525)
	GRASP	0.044 (0.007)	0.043 (0.003)	10.220 (1.533)	17.700 (2.655)
	GIGG	0.048 (0.007)	0.031 (0.005)	13.000 (1.950)	16.400 (2.460)

Table 5: Empirical MSE for non-null coefficients under different simulation settings and correlation structures. Standard deviations across replications are shown in parentheses.

(ρ_w, ρ_b)	Prior	Overall MSE			
		Setting 1	Setting 2	Setting 3	Setting 4
(0.8, 0.2)	BGHS	0.020 (0.002)	0.0212 (0.002)	24.312 (2.431)	14.392 (1.439)
	GRASP	0.018 (0.002)	0.0210 (0.002)	13.344 (1.334)	2.325 (0.233)
	GIGG	0.028 (0.003)	0.0290 (0.003)	13.485 (1.349)	11.620 (1.162)
(0.5, 0.1)	BGHS	0.010 (0.001)	0.0130 (0.001)	28.521 (2.852)	45.000 (4.500)
	GRASP	0.014 (0.001)	0.0100 (0.001)	13.272 (1.327)	108.168 (10.817)
	GIGG	0.016 (0.002)	0.0070 (0.001)	12.050 (1.205)	8.695 (0.870)
(0.3, 0.01)	BGHS	0.006 (0.001)	0.0060 (0.001)	9.701 (0.970)	43.070 (4.307)
	GRASP	0.008 (0.001)	0.0050 (0.001)	9.222 (0.922)	9.203 (0.920)
	GIGG	0.007 (0.001)	0.0060 (0.001)	11.665 (1.167)	8.833 (0.883)

Table 6: Overall empirical MSE under different simulation settings and correlation structures. Standard deviations across replications are shown in parentheses.

Before examining each experimental setting individually, we highlight several consistent patterns that emerge across more or less all the settings :

- GRASP consistently achieves the lowest or comparable MSE across in almost most of the scenarios and correlations. This is because GRASP’s adaptive shrinkage at both the individual and group levels allows it to balance the trade-off between shrinking noise and preserving the signals more effectively than the other methods.
- GIGG performs particularly well in setting 2 (Distributed), where its group shrinkage mechanism ($\delta_g^2 \sim \text{Gamma}(a_g, 1)$) pools information across correlated predictors, making it ideal for distributed signals. However, in concentrated or mixed settings, its performance is less consistent.
- BGHS consistently has the highest MSE across most settings, particularly in dense and high correlation scenarios. Its fixed *half-Cauchy shrinkage* can not adapt to varying sparsity patterns or groups structures, leading to over-shrinkage of signals and under-shrinkage of noise.
- MSE decreases in overall as correlation decreases. This is because lower within-group-correlation reduces multicollinearity, making it easier to distinguish signals from the noise.

Key characteristics of *setting 1 (Concentrated)* is $p = 50$ out of which $p_0 = 45, p_1 = 5$, only one coefficient per group is active. This is a sparse setting where signals are isolated within groups. GRASP achieves the lowest null MSE at highest correlation, demonstrates its ability to aggressively shrink null coefficients. This is due to the beta prime prior. GIGG performs slightly worse because its groups shrinkage may overshrink the isolated signals, leading to a less aggressive shrinkage of nulls within the same group. A similar observation can be given for the non-null MSE also, where GRASP has the lowest which indicates the escape of single coefficient from the shrinkage. As correlation decreases from 0.8 to 0.3, all methods show improved performance, but GRASP maintain its advantage, confirming that adaptive individual shrinkage is most beneficial when signals are sparse within within groups.

In the *distributed* setting, where all predictors in the first group are active, GIGG achieves the lowest null MSE, this is because GIGG’s group shrinkage parameter has group-specific

hyperparameters which pools information across 10 active coefficients in the first group. GRASP also performs well as compared to BGHS, but GIGG overpowers everyone. Even, non-null MSE is lesser for GIGG in this case.

In the *dense setting* (experimental setting 3), where most coefficients are active, GRASP achieves the lowest null MSE and the lowest non-null MSE. Note that, GIGG has similar non-null MSE but higher null MSE, indicating that its group shrinkage is less effective when most coefficients are effective– it may overshrink some signals. BGHS shows a very high total MSE as compared to remaining two.

In the *mixed setting*, where some groups are sparse, and some are entirely null, GRASP achieves the lowest non-null MSE at high correlation, reflecting its ability to adaptively identify and preserve signals in both dense and sparse groups when the group structure is clearly defined. However at moderate correlation and low correlation setup, its non-null MSE is very high. GIGG and GRASP works similarly from the perspective of null MSE but GIGG has lesser non-null MSE than GRASP.

7.2 ROC curves

In this section, the ROC curves are presented along with table comprising the corresponding AUC values. The ROC curve is obtained using the methodology discussed in section 6.2.2. The results are averaged over 100 replications.

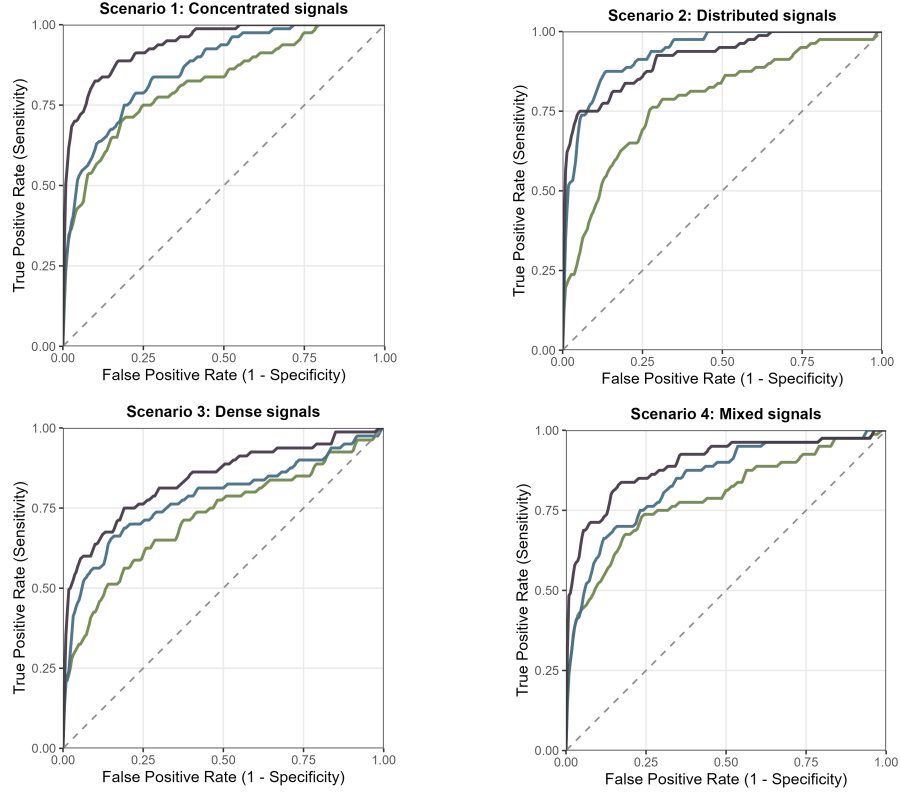
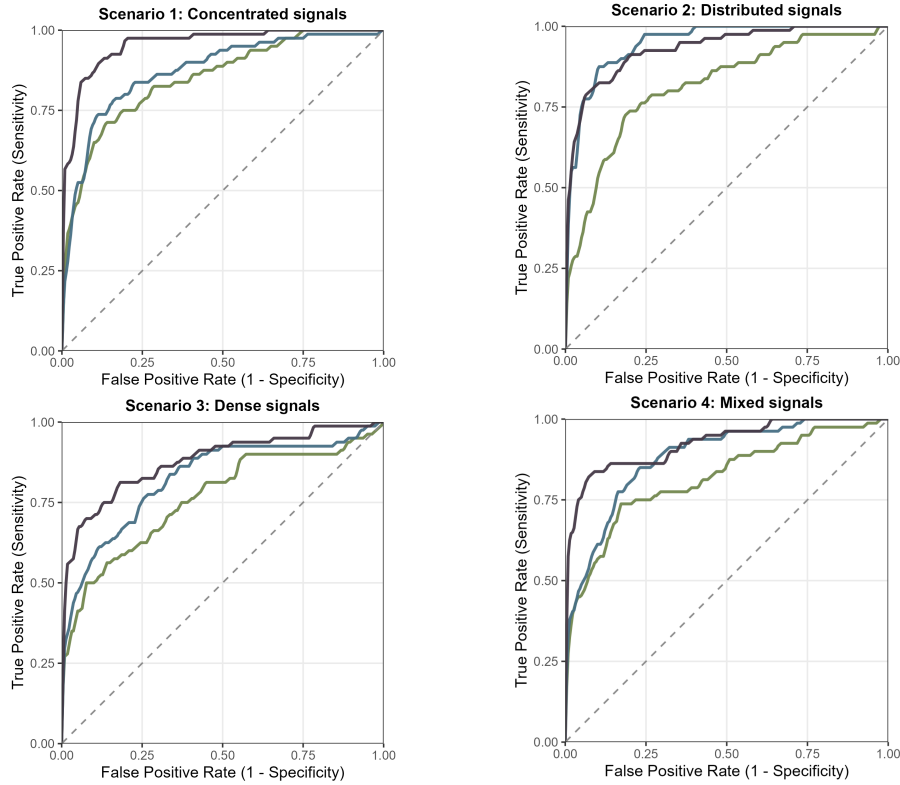
(ρ_w, ρ_b)	Prior	AUC values			
		Setting 1	Setting 2	Setting 3	Setting 4
(0.8, 0.2)	BGHS	0.8180	0.6839	0.7201	0.7895
	GRASP	0.9383	0.8960	0.8408	0.8947
	GIGG	0.8647	0.9341	0.7809	0.8824
(0.5, 0.1)	BGHS	0.8501	0.7142	0.7612	0.8162
	GRASP	0.9548	0.9311	0.8781	0.9243
	GIGG	0.8725	0.9480	0.8237	0.9010
(0.3, 0.01)	BGHS	0.8783	0.8445	0.7987	0.8665
	GRASP	0.9697	0.9053	0.9204	0.9477
	GIGG	0.8921	0.9597	0.8625	0.9302

Table 7: AUC values for BGHS, GRASP, and GIGG across all experimental settings and correlation structures. Values are averaged over 100 replications

The ROC curves are represented in the following three separate figures, one for each correlation structure. Each figure contains a 2×2 grid of panels, with each panel corresponding to one of the four experimental scenarios. The color codes used for each of the method is explained below:

— BGHS — GIGG — GRASP

The grey dashed diagonal line in each figure represents the performance of a random classifier. Curves lying above this diagonal indicate discriminatory ability, superior to random guessing, with higher curves corresponding to better performance.

Figure 4: ROC curves under high within-group correlation $\rho_w = 0.8, \rho_b = 0.2$ Figure 5: ROC curves under moderate within-group correlation $\rho_w = 0.5, \rho_b = 0.1$

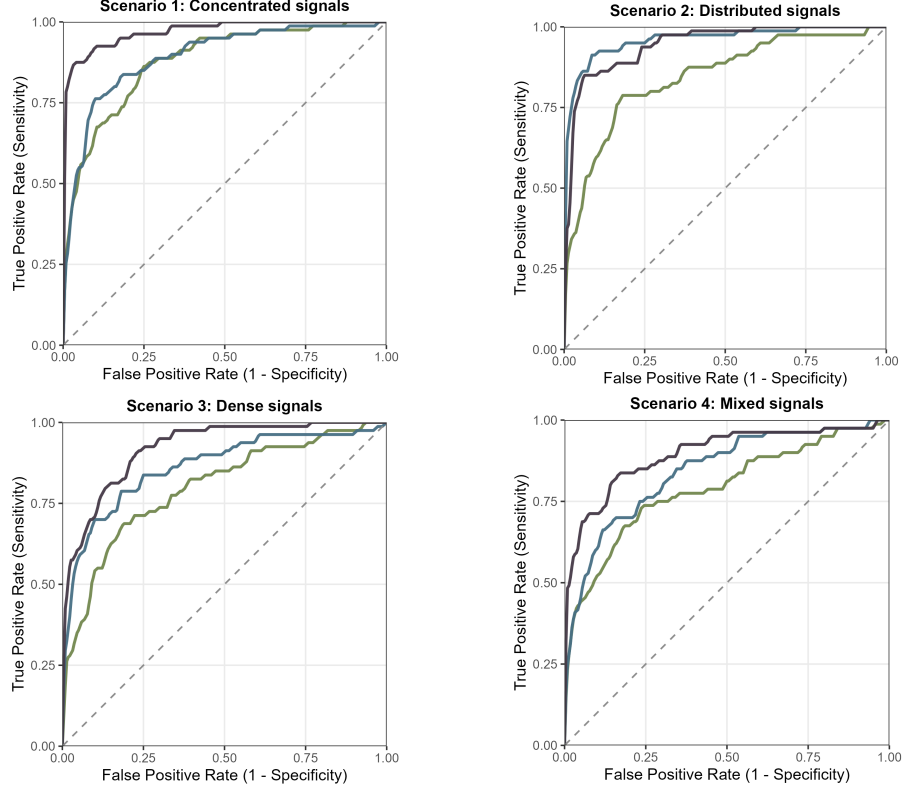


Figure 6: ROC curves under low within-group correlation ($\rho_w = 0.3, \rho_b = 0.01$).

From the above three figures and table 7, we can observe the following things. The results reveals a consistent and theoretically meaningful hierarchy : GRASP consistently achieves the highest AUC, followed by GIGG with BGHS trailing as the lowest performer.

In the *concentrated setting*, GRASP achieves a very high AUC, ranging from 0.938 to 0.969. Beta prime prior helps to escape that *lone active coefficient* and it has included in the model to get the best prediction of the model. But on the other hand, in the *distributed setting*, where all predictors in the first group are active, ROC curve for GIGG is placed higher than the two other ROC curves. GRASP comes closer to GIGG for moderate correlation setup, but in the other two GIGG overpowers every other methods. BGHS's predictive power is very low as compared to the other two.

In the *dense setting*, GRASP still achieves the greatest AUC, followed by GIGG and BGHS. GIGG's group shrinkage is less beneficial for prediction in dense settings, as pooling information across groups may overshrink. In the *mixed setting*, GRASP achieves the highest AUC overall, but here GIGG is giving a tough competition to it. From the table 7, we can observe the AUC values for GIGG is very close to the GRASP. This experimental setting has several groups with several sparsity patterns, so using different hyperparameters for each group (which is done by GIGG) produces better predictions.

Also, we can observe that with the decrease in the within-group and between-group correlation, BGHS's predictive power improves than its previous cases.

Across all the scenarios and correlation structures, GRASP consistently dominates in most of the cases in terms of greater AUC. But specifically, GIGG performs much better than GRASP in experimental setting 2 (Distributed) and also it gives a comparable AUC for mixed signal scenario.

8 Application to real-world data

8.1 Overview

To evaluate the practical performance of the three bayesian grouped shrinkage methods – BGHS, GRASP and GIGG – under logistic regression framework, we apply the proposed methodologies to a real-world bankruptcy prediction dataset. This section provides a detailed description of the dataset, the rationale behind the variable selection, the construction of meaningful groups based on domain knowledge, and the comparative framework adopted for assessing the performance of the three methods.

Bankruptcy prediction is a classical and critically important problem in financial risk management. Accurate identification of firms at risk of financial distress enables creditors, investors and regulatory bodies to take timely preventive actions.

8.2 Data description

The dataset used in this study, sourced from the *Taiwan Economic Journal* for the years 1999-2009¹, comprises financial ratios of the firms, with the binary response variable indicating whether a firm became bankrupt within a specified time frame. The dataset consists of **6819** observations and 95 financial predictors, along with a binary response variable indicating the bankruptcy status of each firm. The predictors represent a comprehensive collection of financial ratios commonly used in credit risk assessment and failure prediction. These ratios can be broadly categorized into several financial dimensions, including *profitability*, *liquidity*, *leverage*, *efficiency*, *cash flow* and *asset structure*. The dataset exhibits a *significant class imbalance* and poses additional challenges for classification.

A basic exploratory data analysis is conducted to understand the structure and characteristics of the data. The EDA included an examination of the response distribution, summary statistics and an assessment of missing values. There is no missing observation in the data. Based on the the domain knowledge, 37 variables are selected for the and suspected to be associated with the response. The column names for 37 selected variables have been shortened for readability and for the ease of reference throughout this section. The following table explains the variables briefly and figure 7 represents the correlation plot of the 37 selected predictors and the predictors in the axes of the plot are taken in the same order as it is in the following table.

Table 8: Description of the 37 Selected Predictors with Group Membership

Group	Short Name	Economic Description
Profitability	NI_to_Assets	Net income relative to total assets; measures overall efficiency in generating profit from assets.
	ROA_A	Return on assets (after tax); indicates how efficiently a firm uses its assets to generate after-tax profit.
	ROA_B	Return on assets (before interest and depreciation after tax); measures operating efficiency.
	ROA_C	Return on assets (before interest and depreciation before interest); measures operational performance.

¹Data source : <https://archive.ics.uci.edu/dataset/572/taiwanese+bankruptcy+prediction>

Group	Short Name	Economic Description
	NPBT_to_Capital	Net profit before tax relative to paid-in capital; measures return on equity capital before tax.
	NPBT_per_Share	Net profit before tax per share; indicates profitability on a per-share basis before tax.
	NI_to_Equity	Net income relative to shareholders' equity; measures return on equity.
	OP_per_Share	Operating profit per share; measures operational efficiency on a per-share basis.
	OP_to_Capital	Operating profit relative to paid-in capital; measures operational return on capital.
	GP_to_Sales	Gross profit as a percentage of sales; measures core profitability after cost of goods sold.
	Op_Gross_Margin	Operating gross margin; measures profitability after operating expenses.
	Realized_GM	Realized sales gross margin; measures actual gross profit realised from sales.
	OP_per_Person	Operating profit per employee; measures employee productivity and operational efficiency.
Cash Flow	CF0_to_Assets	Cash flow from operations relative to total assets; measures cash-generating efficiency.
	Cash_to_Assets	Cash holdings relative to total assets; measures liquidity and financial flexibility.
	Cash_to_CL	Cash relative to current liabilities; measures short-term solvency.
	CF_per_Share	Cash flow per share; measures cash generation on a per-share basis.
Per Share	Persistent_EPS	Persistent earnings per share over four seasons; measures consistent profitability.
	NVS_A	Net value per share (A); measures book value per share.
	NVS_B	Net value per share (B); alternative measure of book value per share.
	NVS_C	Net value per share (C); additional measure of book value per share.
Leverage	Debt_Ratio	Total debt as a percentage of total assets; measures financial leverage and risk.
	Borrowing_Dep	Dependence on borrowing; measures reliance on external debt financing.
Asset Structure	NW_to_Assets	Net worth relative to total assets; measures equity cushion and financial stability.
	RE_to_Assets	Retained earnings relative to total assets; measures accumulated profitability.
	CL_to_Assets	Current liabilities relative to total assets; measures short-term debt burden.
	WC_to_Assets	Working capital relative to total assets; measures short-term financial health.
	CL_to_CA	Current liabilities relative to current assets; measures short-term solvency.

Group	Short Name	Economic Description
	Liability_to_Equity	Total liabilities relative to equity; measures financial leverage.
	CL_to_Equity	Current liabilities relative to equity; measures short-term debt burden on equity.
	WC_to_Equity	Working capital relative to equity; measures liquidity relative to shareholder investment.
	Expense_to_Assets	Total expenses relative to total assets; measures cost efficiency.
	Equity_to_LTL	Equity relative to long-term liabilities; measures long-term solvency.
	QA_to_Assets	Quick assets relative to total assets; measures liquid asset holdings.
	Equity_to_Liability	Equity relative to total liabilities; inverse measure of financial leverage.
Other (Tax and liability)	CL_to_Equity_2	Current liabilities to equity (alternative measure); short-term debt burden.
	Tax_Rate	Effective tax rate (A); measures tax burden on profits.

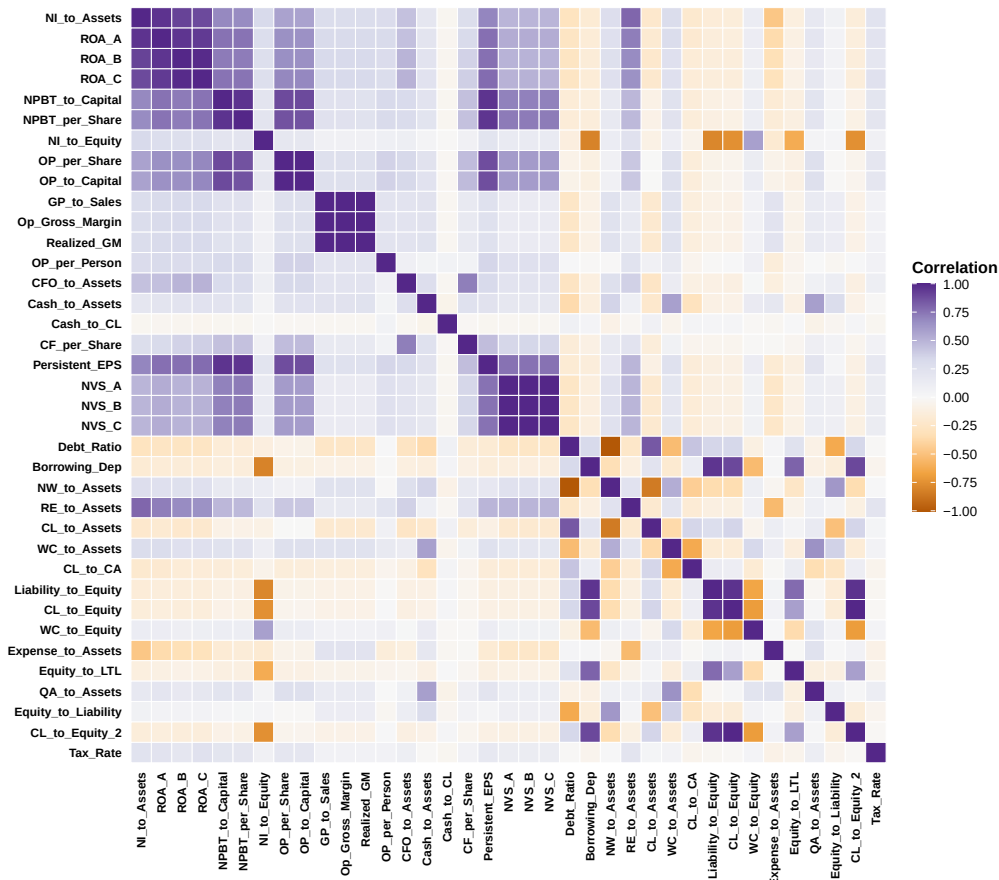


Figure 7: Correlation plot of the 37 predictors

The aforementioned grouping structure also can be observed from the above figure.

8.3 Results and interpretation

To evaluate the predictive performance of the three bayesian grouped shrinkage methods on unseen data, we conducted a 70%-30% train-test split, i.e. 4773 observations in the training set and remaining in the test set. The dataset is standardized before the analysis. In the whole data, there is a class imbalance with only 220 bankrupt firms which is about 3.3%.

8.3.1 Credible interval analysis and confusion matrix

The 95% credible intervals are shown in the figure 8. GIGG consistently produces the *narrowest* CI within the groups exhibiting dense signal structures. This is most pronounced in *profitability* and *leverage* group, where GIGG's intervals are almost 10-20% narrower. The narrowing is most pronounced for NI_to_Asset and ROA_A. This confirms the theoretical advantage of GIGG's

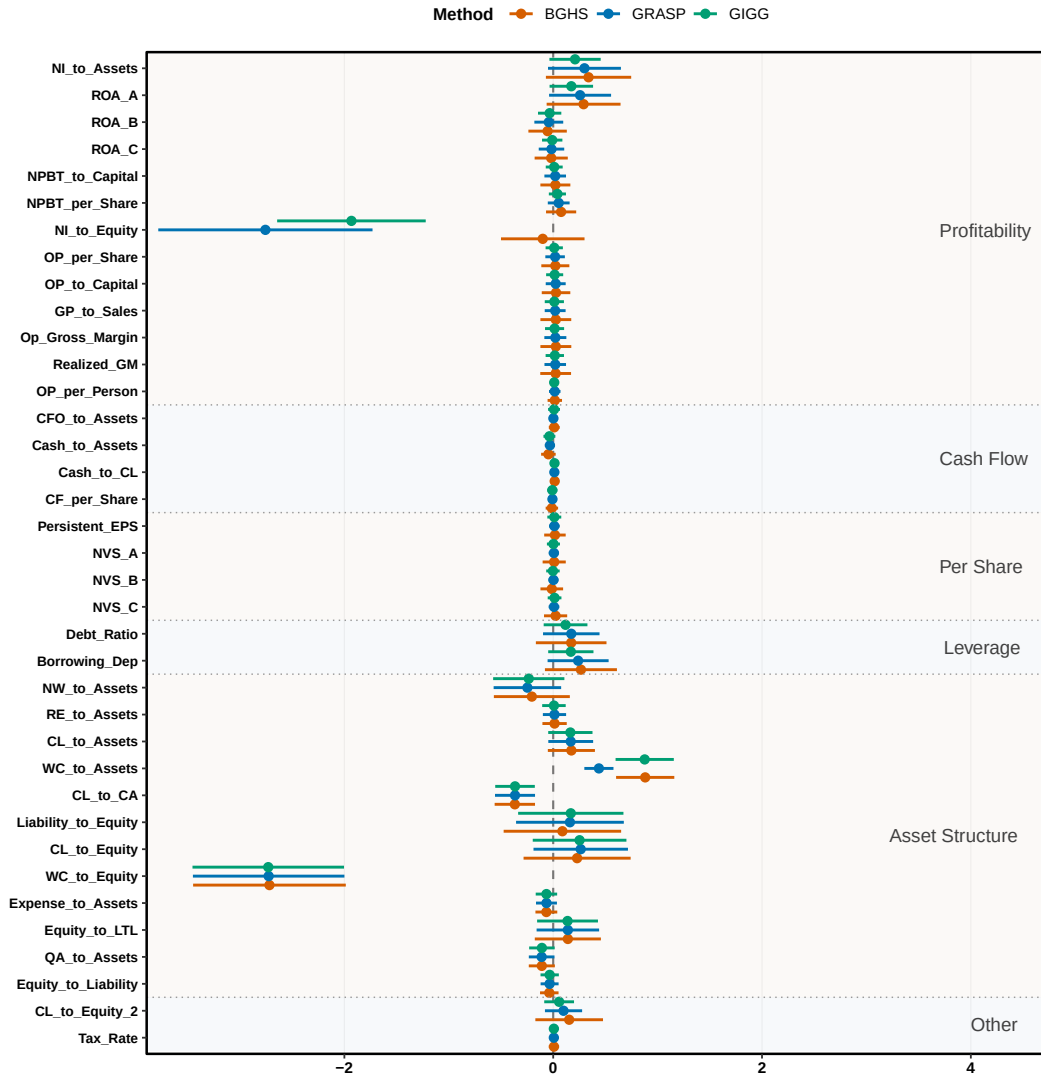


Figure 8: Confidence intervals for each regression coefficients

group inverse gamma prior. When variables within a group are heavily correlated (which is here for the Profitability group) and have similar effects then group-level shrinkage parameter effectively borrows strength across the group, reducing estimation variance. For NI_to_Equity, BGHS produces a CI that contains zero (-0.500, 0.300), meaning BGHS fails to select this variable. In contrast, both GRASP and GIGG produce CIs that do not contain zero, correctly identifying NI_to_Equity as an important predictor. This is a striking example of how adaptive methods outperform non-adaptive shrinkage in variable selection.

The *Cash Flow* group measures a firm’s ability to generate cash and meet short-term obligations. This group exhibits sparse signal structure, where one variable (CFO_to_Assets) is a strong isolated predictor. Here, GRASP emerges with narrower CIs, but GIGG CI lengths do not differ that much. A similar argument can be drawn for the Per Share group, but there BGHS CI’s comparatively much wider than the other two.

GIGG produces the narrowest credible intervals for both *Leverage* variables. The narrowing is substantial: for Debt_Ratio, GIGG’s interval is approximately 35% narrower than BGHS’s; for Borrowing_Dep, it is approximately 37% narrower.

The *Asset Structure* group contains variables describing a firm’s asset composition. This group exhibits mixed signal structure with some strong predictors (WC_to_Assets, CL_to_CA) and many weaker ones. The Asset Structure group shows a mixed pattern. For WC_to_Assets, GIGG produces a narrower interval than GRASP (0.597-1.155 vs. 0.299-0.577). For CL_to_CA and WC_to_Equity, all three methods produce similar intervals. This suggests that the signal structure within Asset Structure is heterogeneous: some variables are strong and dense (benefiting from GIGG), while others are more isolated (benefiting from GRASP).

GIGG produces the narrowest credible intervals for both variables in the last group tax and liability (shown as *Other* in the plot). The pattern aligns with the Leverage group, suggesting that these variables benefit from group-level shrinkage despite their small group size.

Predicted Class	Actual Class		Method
	Non-Bankrupt (0)	Bankrupt (1)	
Non-Bankrupt (0)	1775	9	BGHS
Bankrupt (1)	217	45	
Non-Bankrupt (0)	1772	16	GRASP
Bankrupt (1)	193	65	
Non-Bankrupt (0)	1775	13	GIGG
Bankrupt (1)	186	72	

Table 9: Confusion matrices for BGHS, GRASP, and GIGG on the Taiwan bankruptcy dataset

In the test set, there are 1974 no-bankrupt firms and remaining are bankrupt firms. All three methods have achieved high accuracy, driven primarily by high specificity. This indicates

that the models are excellent at correctly identifying non-bankrupt firms - it minimizes the false alarms. However, the low sensitivity (17-28%) across all methods highlights the fundamental challenge of bankruptcy prediction with severely imbalanced data. But as we compare we observe that GRASP and GIGG has identified correctly 20 and 27 more firms respectively than BGHS.

BGHS has lowest sensitivity and most false negatives (217), where as highest sensitivity is for GIGG. Also, BGHS is overly conservating; rarely predicts bankruptcy (only in 45 correct + 9 false = 54 cases), it misses most bankrupts firm.

8.3.2 Insights from the variable selection patterns

Before presenting the specific variables selected by each method in table 10, it is worth noting that each of the three Bayesian grouped regression approaches perceives the bankruptcy problem from a fundamentally different perspective. The variables selected by a method reveal not only which financial factors are associated with bankruptcy risk, but also how *that method's prior structure shapes its understanding of financial distress*.

The *BGHS method* selects only four variables, all of which measure the firm's liquidity and short-term financial health, while completely ignoring profitability and leverage. Cash_to_Assets measures the cash buffer a company holds relative to its total size, and when this is low, the firm lacks the emergency funds needed to survive unexpected expenses or economic shocks. CL_to_CA (Current Liabilities to Current Assets) compares short-term debts to short-term assets, and a high value indicates the company owes more than it can immediately pay, directly signaling imminent solvency risk. WC_to_Equity (Working Capital to Equity) measures liquid assets relative to shareholder investment, and when negative, it means the company is using investor money to fund daily operations, a clear distress signal. WC_to_Assets captures the working capital cushion relative to the firm's total size, reflecting its ability to comfortably meet short-term obligations.

BGHS's fixed half-Cauchy shrinkage is overly aggressive on the profitability group, shrinking all those signals toward zero, which explains why no profitability variables are selected. The method essentially concludes that bankruptcy risk for Taiwanese firms is primarily a story of **liquidity failure rather than profitability decline**.

Group	BGHS	GRASP	GIGG
Profitability	0	1 (7)	2 (12, 13)
Cash Flow	1 (2)	0	1 (2)
Per Share	0	1 (4)	0
Leverage	0	0	0
Asset Structure	3 (4, 5, 8)	3 (4, 5, 8)	3 (5, 8, 11)
Other	0	0	0
Total	4	5	6

Note: Numbers in parentheses indicate the corresponding variable number within each predictor group, as defined in Table 8. That is, (2) means 2nd variable of that respective group.

Table 10: Selected variables within each predictor group under each methods for the Taiwan bankruptcy data

GRASP selects five variables, combining liquidity measures with profitability and asset valuation, reflecting its adaptive individual shrinkage that allows important isolated signals to "escape" group-level regularization. *NI_to_Equity* (Net Income to Equity) measures return on shareholder investment, and when negative, it indicates the company is destroying shareholder value and depleting its capital base, making eventual failure inevitable. *WC_to_Assets*, *CL_to_CA*, and *WC_to_Equity* are the same liquidity measures selected by *BGHS*, confirming that short-term solvency is universally important. Additionally, *NVS_C* (Net Value per Share) captures the book value per share, and when this is low, it suggests the company's assets have limited liquidation value, providing little cushion for creditors if the firm fails. *GRASP*'s ability to select *NI_to_Equity* while *BGHS* cannot demonstrates the advantage of adaptive shrinkage, as the method allows profitability signals to emerge when they are genuinely important.

The selection of *NVS_C* is particularly interesting because the three *NVS* variables (*A*, *B*, *C*) are highly correlated, and *GRASP* intelligently identifies the strongest signal (*NVS_C*) while shrinking the redundant ones. The economic story from *GRASP* is that bankruptcy risk stems from **a combination of profitability failure** (*NI_to_Equity*), **liquidity crisis** (*WC_to_Assets*, *CL_to_CA*, *WC_to_Equity*), and **depleted asset value** (*NVS_C*), presenting a more complete risk assessment than *BGHS*.

GIGG selects the most variables (six), providing the most comprehensive picture by capturing core profitability, employee efficiency, liquidity, and asset quality, reflecting its group-specific hyperparameters that allow each group to contribute what matters most. *Realized_GM* (Realized Gross Margin) measures the true profit the company makes on its core business operations after accounting for costs, and when this margin is thin or negative, the firm operates on a razor's edge with no buffer for mistakes or downturns. *OP_per_Person* (Operating Profit per Employee) captures employee productivity, and low values indicate inefficient operations where labor costs are not justified by output, a clear sign of organizational distress. *Cash_to_Assets* measures the cash buffer, and *CL_to_CA* captures short-term solvency, while *WC_to_Equity* reflects the relationship between liquidity and capital structure, all three confirming the importance of liquidity management. Finally, *QA_to_Assets* (Quick Assets to Assets) provides a stricter, more conservative liquidity measure by excluding inventory, recognizing that in times of distress, inventory may be unsellable at book value, so only cash, marketable securities, and receivables provide reliable emergency funds.

GIGG's advantage lies in its group-specific parameters, which allow the Profitability group to contribute two variables (*Realized_GM* and *OP_per_Person*) while *BGHS* and *GRASP* each contribute only one or none. The economic narrative from *GIGG* is that bankruptcy is a multi-faceted phenomenon involving **thin margins** (*Realized_GM*), **poor workforce efficiency** (*OP_per_Person*), **inadequate cash buffers** (*Cash_to_Assets*, *QA_to_Assets*), **excessive short-term obligations** (*CL_to_CA*), and **depleted equity cushions** (*WC_to_Equity*), providing the most complete and nuanced understanding of financial distress.

9 Discussion

This dissertation set out to extend the idea of Bayesian grouped regression to the logistic regression setup, addressing the fundamental challenge posed by the lack of conjugacy in the logistic likelihood. The Pólya-Gamma data augmentation framework of Polson et al. (2013) proved to be a powerful and elegant solution, transforming the intractable logistic likelihood into a conditionally Gaussian form that enabled efficient posterior inference via Gibbs sampling. This augmentation strategy formed the backbone of our methodological development, allowing us to derive full conditional posterior distributions for three prominent grouped shrinkage priors—the Bayesian Grouped Horseshoe (BGHS), the Grouped Regression with Adaptive Shrinkage Prior (GRASP), and the Group Inverse-Gamma Gamma (GIGG) prior—and implement them in a unified computational framework. The ability to restore conjugacy through this augmentation not only simplified computation but also opened the door for applying these sophisticated shrinkage priors to binary response settings, which had remained largely unexplored in the existing literature.

The simulation study provided valuable insights into the comparative performance of the three methods across a variety of signal configurations and correlation structures. GRASP emerged as the most consistently reliable method, achieving the lowest mean squared error across most experimental settings and delivering the highest area under the ROC curve in predictive performance. This superiority can be attributed to its adaptive shrinkage mechanism, which employs Beta-prime priors on both local and group-level shrinkage parameters, allowing it to balance the competing demands of shrinking noise and preserving substantial signals effectively. GIGG, on the other hand, demonstrated exceptional performance in settings where signals were distributed across multiple predictors within a group, such as the distributed experimental setting, where all predictors in the first group were active.

This strength stems from its group-specific hyperparameters, which enable effective pooling of information across correlated predictors within a group. However, GIGG’s performance was less consistent in concentrated or mixed signal settings, where its group shrinkage mechanism tended to overshrink isolated signals. BGHS, while serving as a useful baseline, consistently underperformed relative to the other two methods, highlighting the limitations of its fixed half-Cauchy shrinkage in adapting to varying sparsity patterns and group structures. The simulation results collectively underscore the importance of adaptive shrinkage in grouped regression settings and demonstrate that the choice of prior can have a substantial impact on both estimation accuracy and predictive performance.

The application to the *Taiwan bankruptcy dataset* revealed how each method perceives bankruptcy from a distinct perspective. BGHS selected only liquidity-based variables, concluding that bankruptcy is primarily a story of liquidity failure. GRASP combined liquidity measures with profitability and asset valuation, presenting a more balanced view. GIGG selected the most variables, capturing profitability, employee efficiency, liquidity, and asset quality. The credible interval analysis confirmed GIGG’s advantage in producing the narrowest intervals for dense signal groups, while the confusion matrices showed that all methods struggled with sensitivity due to class imbalance, though GIGG correctly identified the most bankrupt firms.

This application demonstrated that the choice of prior shapes both statistical performance and economic interpretation of the model.

In conclusion, this work has successfully extended three important grouped shrinkage priors to the logistic regression framework, providing a comprehensive comparison of their performance through both simulation and real-data analysis. The findings suggest that GRASP offers the best overall performance for sparse and mixed signal settings, while GIGG is particularly well-suited for scenarios with dense or distributed signals within groups. BGHS, despite its limitations, remains a useful baseline for situations where prior information is weak or computational simplicity is prioritized. The methodological framework developed in this dissertation provides a solid foundation for future research, including extensions to other generalized linear models, incorporation of overlapping group structures, and the development of more computationally efficient algorithms for high-dimensional settings.

10 Future scopes

While this study provides a comprehensive comparison of BGHS, GRASP and GIGG under logistic regression, several promising directions remain underexplored.

First, this study compares the *full Bayesian setup* of these priors. An alternative approach would be to estimate the hyperparameters by maximum marginal likelihood estimates (MMLE), potentially offering computational advantages and a different perspective on prior tuning.

Second, from the analysis, we observed that BGHS mainly suffers from *non-adaptiveness and heavy shrinkage*. One promising direction is to incorporate prior information—for instance, from previous studies or domain knowledge—or to otherwise improve the adaptability of BGHS. Research has shown that such prior-informed approaches can be effective in high-dimensional sparse regression settings.

Third, this project extends GRASP and GIGG only to logistic regression. A natural extension would be to generalize these priors to other members of the exponential family, such as Poisson regression for count data, multinomial regression for multi-class outcomes, and negative binomial regression for overdispersed counts.

While GIGG has established posterior consistency and concentration results for linear regression models, these theoretical guarantees have not yet been evaluated for logistic regression or other non-normal outcomes. Investigating posterior contraction rates, variable selection consistency, and optimal minimax rates for GIGG and GRASP in the logistic regression framework would provide a firmer theoretical foundation for their empirical success and offer deeper insights into their behavior under various sparsity regimes. Such theoretical developments would also help guide the choice of hyperparameters and establish conditions under which these priors achieve optimal performance in binary response settings.

11 Code repository and R package

All R codes developed for this dissertation are publicly available on GitHub. That GitHub repository can be accessed through :

[https://github.com/TathastuGhosh/
Extending-the-idea-of-Bayesian-Grouped-Regression-under-Logistic-Setup](https://github.com/TathastuGhosh/Extending-the-idea-of-Bayesian-Grouped-Regression-under-Logistic-Setup)

Additionally, an R package named

`bayesglogit`

has been developed to facilitate the application of these methods. The package is hosted on the following GitHub profile

<https://github.com/TathastuGhosh>

within a dedicated repository. Its `README` file contains complete instructions for installation and usage.

Appendix

Appendix A Distributional Forms and Parametric Representations

Normal distribution : If $X \sim \mathcal{N}(\mu, \sigma^2)$, then its density is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}, \quad x \in \mathbb{R}.$$

with mean μ , variance σ^2 and MGF $M_X(t) = \exp \left\{ \mu t + \frac{1}{2} \sigma^2 t^2 \right\}$

Multivariate Normal Distribution : If $\mathbf{X}^{k \times 1} \sim \mathcal{N}_k(\boldsymbol{\mu}, \Sigma)$, then its density

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{k/2} \sqrt{|\Sigma|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}, \quad \boldsymbol{\mu} \in \mathbb{R}^k, \Sigma \text{ pd}$$

with MGF $M_{\mathbf{X}}(\mathbf{t}) = \exp \left\{ \mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t} \right\}$

Gamma distribution : If $X \sim \text{Gamma}(a, b)$, having shape parameter $a > 0$ and rate parameter $b > 0$, then

$$f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}, \quad x > 0.$$

with mean $\frac{a}{b}$ and variance $\frac{a}{b^2}$.

Inverse-Gamma distribution : If $X \sim \text{IG}(a, b)$, then

$$f(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} e^{-b/x}, \quad x > 0.$$

with mean $\frac{b}{a-1}, a > 1$ and variance $\frac{b^2}{(a-1)^2(a-2)}, a > 2$.

Generalized inverse Gaussian distribution : If $X \sim \text{GIG}(\lambda, \chi, \psi)$, then

$$f(x) = \frac{(\psi/\chi)^{\lambda/2}}{2K_\lambda(\sqrt{\chi\psi})} x^{\lambda-1} \exp \left\{ -\frac{1}{2} (\chi x^{-1} + \psi x) \right\}, \quad x > 0.$$

where $\psi > 0, \chi > 0, \lambda \in \mathbb{R}$ and $K_\lambda(\cdot)$ is the modified third kind Bessel function with index λ .

Half Cauchy distribution : If $X \sim \mathcal{C}^+(\mu, \sigma)$, then

$$f(x) = \frac{2}{\pi\sigma} \frac{1}{1 + (y - \mu)^2/\sigma^2}, \quad x \geq \mu.$$

which is basically the truncated Cauchy distribution on the interval greater than equals to the peak.

Beta prime distribution : If $X \sim \mathcal{B}'(a, b)$ having shape parameter $a, b > 0$, then

$$f(x) = \frac{x^{a-1}(1+x)^{-a-b}}{B(a, b)}, \quad x > 0$$

with mean $\frac{a}{b-1}$ and variance $\frac{a(a+b-1)}{(b-2)(b-1)^2}$.

Appendix B Full Conditional Posterior Distributions

This section will produce the structural form of each of the full conditional posterior distribution from the joint posterior distribution. The posterior distribution will be obtained for each model as it is described in the previous sections.

All notations used here are explained previously in detail.

B.1 BGHS model

Joint posterior distribution is given by :

$$\begin{aligned} & \propto \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} - \frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \cdot \prod_{g=1}^G \prod_{j=1}^{p_g} \frac{1}{\tau \lambda_{gj} \delta_g} \exp \left\{ -\frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2} \right\} \\ & \cdot \prod_{g=1}^G \prod_{j=1}^{p_g} \left\{ \frac{1}{\sqrt{t_{gj}} \Gamma(\frac{1}{2})} (\lambda_{gj}^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{t_{gj} \lambda_{gj}^2} \right\} \frac{1}{\Gamma(\frac{1}{2})} (t_{gj})^{-\frac{3}{2}} \exp \left\{ -\frac{1}{t_{gj}} \right\} \right\} \\ & \cdot \prod_{g=1}^G \left\{ \frac{1}{\sqrt{c_g} \Gamma(\frac{1}{2})} (\delta_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{c_g \delta_g^2} \right\} \frac{1}{\Gamma(\frac{1}{2})} (c_g)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{c_g} \right\} \right\} \\ & \cdot (\tau^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{\nu \tau^2} \right\} \frac{1}{\sqrt{\nu} \Gamma(\frac{1}{2})} \cdot \frac{1}{\Gamma(\frac{1}{2})} \nu^{-\frac{3}{2}} \exp \left\{ -\frac{1}{\nu} \right\} \end{aligned}$$

Then the full conditional posterior distributions are :

$$p(\omega_i | \cdot) \propto \exp \left\{ -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i)$$

$$p(\boldsymbol{\beta} | \cdot) \propto \exp \left\{ \boldsymbol{\kappa}^\top X \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top (X^\top \Omega X) \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top D^{-1} \boldsymbol{\beta} \right\}$$

$$\begin{aligned}
p(\lambda_{gj}^2 \mid \cdot) &\propto (\lambda_{gj}^2)^{-2} \exp \left\{ -\frac{1}{\lambda_{gj}^2} \left(\frac{1}{t_{gj}} + \frac{\beta_{gj}^2}{2\tau^2\delta_g^2} \right) \right\} \\
p(t_{gj} \mid \cdot) &\propto (t_{gj})^{-2} \exp \left\{ -\frac{1}{t_{gj}\lambda_{gj}^2} \right\} \exp \left\{ -\frac{1}{t_{gj}} \right\} \\
p(\delta_g^2 \mid \cdot) &\propto (\delta_g^2)^{-\left(\frac{p_g+1}{2}+1\right)} \exp \left\{ -\frac{1}{\delta_g^2} \left(\frac{1}{c_g} + \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2\lambda_{gj}^2} \right) \right\} \\
p(c_g \mid \cdot) &\propto (c_g)^{-2} \exp \left\{ -\frac{1}{c_g\delta_g^2} \right\} \exp \left\{ -\frac{1}{c_g} \right\} \\
p(\tau^2 \mid \cdot) &\propto (\tau^2)^{-\left(\frac{p+3}{2}\right)} \exp \left\{ -\frac{1}{\tau^2} \sum_{g=1}^G \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\delta_g^2\lambda_{gj}^2} \right\} \exp \left\{ -\frac{1}{\nu\tau^2} \right\} \\
p(\nu \mid \cdot) &\propto \nu^{-2} \exp \left\{ -\frac{1}{\nu\tau^2} \right\} \exp \left\{ -\frac{1}{\nu} \right\}
\end{aligned}$$

B.2 GRASP model

Joint posterior distribution is given by :

$$\begin{aligned}
&\propto \prod_{i=1}^n \exp \left\{ \kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} - \frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \cdot \prod_{g=1}^G \prod_{j=1}^{p_g} \frac{1}{\tau \lambda_{gj} \delta_g} \exp \left\{ -\frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2} \right\} \cdot \\
&\prod_{g=1}^G \prod_{j=1}^{p_g} \frac{1}{t_{gj}^{b_g} \Gamma(b_g)} (\lambda_{gj}^2)^{-b_g-1} \exp \left\{ -\frac{1}{t_{gj} \lambda_{gj}^2} \right\} \frac{1}{\Gamma(a_g)} (t_{gj})^{-a_g-1} \exp \left\{ -\frac{1}{t_{gj}} \right\} \\
&\prod_{g=1}^G \frac{1}{c_g^b \Gamma(b)} (\delta_g^2)^{-b-1} \exp \left\{ -\frac{1}{c_g \delta_g^2} \right\} \frac{1}{\Gamma(a)} (c_g)^{-a-1} \exp \left\{ -\frac{1}{c_g} \right\} \\
&\frac{1}{\sqrt{\nu} \cdot \Gamma\left(\frac{1}{2}\right)} (\tau^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{\nu\tau^2} \right\} \frac{1}{\Gamma\left(\frac{1}{2}\right)} \nu^{-\frac{3}{2}} \exp \left\{ -\frac{1}{\nu} \right\} \\
&\prod_{g=1}^G \frac{1}{\sqrt{d_g} \Gamma\left(\frac{1}{2}\right)} (a_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{d_g a_g^2} \right\} \frac{1}{\Gamma\left(\frac{1}{2}\right)} (d_g)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{d_g} \right\} \cdot \\
&\prod_{g=1}^G \frac{1}{\sqrt{e_g} \Gamma\left(\frac{1}{2}\right)} (b_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{e_g b_g^2} \right\} \frac{1}{\Gamma\left(\frac{1}{2}\right)} e_g^{-\frac{3}{2}} \exp \left\{ -\frac{1}{e_g} \right\} \\
&\frac{1}{\sqrt{d} \Gamma\left(\frac{1}{2}\right)} (a^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{da^2} \right\} \frac{1}{\Gamma\left(\frac{1}{2}\right)} (d)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{d} \right\} \\
&\frac{1}{\sqrt{e} \Gamma\left(\frac{1}{2}\right)} (b^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{eb^2} \right\} \frac{1}{\Gamma\left(\frac{1}{2}\right)} (e)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{e} \right\}
\end{aligned}$$

Then the full conditional posterior distributions are :

$$\begin{aligned}
p(\omega_i | \cdot) &\propto \exp \left\{ -\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2 \right\} p(\omega_i) \\
p(\boldsymbol{\beta} | \cdot) &\propto \exp \left\{ \boldsymbol{\kappa}^\top X \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top (X^\top \Omega X) \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top D^{-1} \boldsymbol{\beta} \right\} \\
p(\lambda_{gj}^2 | \cdot) &\propto (\lambda_{gj}^2)^{-(b_g + \frac{1}{2})-1} \exp \left\{ -\frac{1}{\lambda_{gj}^2} \left(\frac{1}{t_{gj}} + \frac{\beta_{gj}^2}{2\tau^2 \delta_g^2} \right) \right\} \\
p(t_{gj} | \cdot) &\propto (t_{gj})^{-(b_g + a_g)-1} \exp \left\{ -\frac{1}{t_{gj} \lambda_{gj}^2} \right\} \exp \left\{ -\frac{1}{t_{gj}} \right\} \\
p(\delta_g^2 | \cdot) &\propto (\delta_g^2)^{-(b + \frac{p_g}{2})-1} \exp \left\{ -\sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2} \right\} \exp \left\{ -\frac{1}{c_g \delta_g^2} \right\} \\
p(c_g | \cdot) &\propto (c_g)^{-(a+b)-1} \exp \left\{ -\frac{1}{c_g \delta_g^2} \right\} \exp \left\{ -\frac{1}{c_g} \right\} \\
p(\tau^2 | \cdot) &\propto (\tau^2)^{-\frac{p}{2}} \exp \left\{ -\sum_{g=1}^G \sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2} \right\} (\tau^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{\nu \tau^2} \right\} \\
p(\nu | \cdot) &\propto \nu^{-2} \exp \left\{ -\frac{1}{\nu \tau^2} \right\} \exp \left\{ -\frac{1}{\nu} \right\} \\
p(a_g^2 | \cdot) &\propto \left(\frac{1}{\Gamma(a_g)} \right)^{p_g} \left\{ \prod_{j=1}^{p_g} t_{gj} \right\}^{-a_g-1} (a_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{d_g a_g^2} \right\} \\
p(d_g | \cdot) &\propto (d_g)^{-2} \exp \left\{ -\frac{1}{d_g a_g^2} \right\} \exp \left\{ -\frac{1}{d_g} \right\} \\
p(b_g^2 | \cdot) &\propto \prod_{j=1}^{p_g} \left\{ \frac{1}{t_{gj}^{b_g} \Gamma(b_g)} (\lambda_{gj}^2)^{-b_g-1} \right\} (b_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{e_g b_g^2} \right\} \\
p(e_g | \cdot) &\propto (e_g)^{-2} \exp \left\{ -\frac{1}{e_g b_g^2} \right\} \exp \left\{ -\frac{1}{e_g} \right\} \\
p(a^2 | \cdot) &\propto \left(\frac{1}{\Gamma(a)} \right)^G \left(\prod_{g=1}^G c_g \right)^{-a-1} (a^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{da^2} \right\} \\
p(d | \cdot) &\propto d^{-2} \exp \left\{ -\frac{1}{da^2} \right\} \exp \left\{ -\frac{1}{d} \right\}
\end{aligned}$$

$$p(b^2 | \cdot) \propto \left(\frac{1}{\Gamma(b)}\right)^G \left(\prod_{g=1}^G \delta_g^2\right)^{-b-1} (b^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{eb^2}\right\} \prod_{g=1}^G \left(\frac{1}{c_g}\right)^b$$

$$p(e | \cdot) \propto e^{-2} \exp\left\{-\frac{1}{eb^2}\right\} \exp\left\{-\frac{1}{e}\right\}$$

B.3 GIGG model

Joint posterior distribution is given by :

$$\propto \prod_{i=1}^n \exp\left\{\kappa_i \mathbf{x}_i^\top \boldsymbol{\beta} - \frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2\right\} p(\omega_i) \cdot \prod_{g=1}^G \prod_{j=1}^{p_g} \frac{1}{\tau \lambda_{gj} \delta_g} \exp\left\{-\frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2}\right\} \cdot$$

$$\prod_{g=1}^G \prod_{j=1}^{p_g} \frac{1}{\Gamma(b_g)} (\lambda_{gj}^2)^{-b_g-1} \exp\left\{-\frac{1}{\lambda_{gj}^2}\right\} \cdot \prod_{g=1}^G \frac{1}{\Gamma(a_g)} (\delta_g^2)^{a_g-1} \exp\{-\delta_g^2\} \cdot$$

$$\frac{1}{\sqrt{\nu} \Gamma(\frac{1}{2})} (\tau^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{\nu \tau^2}\right\} \cdot \frac{1}{\Gamma(\frac{1}{2})} \nu^{-\frac{3}{2}} \exp\left\{-\frac{1}{\nu}\right\} \cdot$$

$$\prod_{g=1}^G \frac{1}{\sqrt{d_g} \Gamma(\frac{1}{2})} (a_g^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{d_g a_g^2}\right\} \cdot \frac{1}{\Gamma(\frac{1}{2})} d_g^{-\frac{3}{2}} \exp\left\{-\frac{1}{d_g}\right\} \cdot$$

$$\prod_{g=1}^G \frac{1}{\sqrt{e_g} \Gamma(\frac{1}{2})} (b_g^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{e_g b_g^2}\right\} \cdot \frac{1}{\Gamma(\frac{1}{2})} e_g^{-\frac{3}{2}} \exp\left\{-\frac{1}{e_g}\right\} \cdot$$

Then the full conditional posterior distributions are :

$$p(\omega_i | \cdot) \propto \exp\left\{-\frac{\omega_i}{2} (\mathbf{x}_i^\top \boldsymbol{\beta})^2\right\} p(\omega_i)$$

$$p(\boldsymbol{\beta} | \cdot) \propto \exp\left\{\boldsymbol{\kappa}^\top X \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top (X^\top \Omega X) \boldsymbol{\beta} - \frac{1}{2} \boldsymbol{\beta}^\top D^{-1} \boldsymbol{\beta}\right\}$$

$$p(\lambda_{gj}^2 | \cdot) \propto (\lambda_{gj}^2)^{-\frac{1}{2}} \exp\left\{-\frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2}\right\} (\lambda_{gj}^2)^{-b_g-1} \exp\left\{-\frac{1}{\lambda_{gj}^2}\right\}$$

$$p(\delta_g^2 | \cdot) \propto (\delta_g^2)^{a_g - \frac{p_g}{2} - 1} \exp\left\{-\sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2}\right\} \exp\{-\delta_g^2\}$$

$$p(\tau^2 | \cdot) \propto (\tau^2)^{-\frac{p}{2}} \exp\left\{-\sum_{j=1}^{p_g} \frac{\beta_{gj}^2}{2\tau^2 \lambda_{gj}^2 \delta_g^2}\right\} (\tau^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{\nu \tau^2}\right\}$$

$$p(\nu | \cdot) \propto (\nu)^{-\frac{1}{2}} \exp\left\{-\frac{1}{\nu \tau^2}\right\} (\nu)^{-3/2} \exp\left\{-\frac{1}{\nu}\right\}$$

$$p(a_g^2 | \cdot) \propto (\delta_g^2)^{a_g-1} \cdot \frac{1}{\Gamma(a_g)} (a_g^2)^{-\frac{3}{2}} \exp\left\{-\frac{1}{d_g a_g^2}\right\}$$

$$p(d_g | \cdot) \propto (d_g)^{-2} \exp \left\{ -\frac{1}{d_g} \left(1 + \frac{1}{a_g^2} \right) \right\}$$

$$p(b_g^2 | \cdot) \propto \left(\frac{1}{\Gamma(b_g)} \right)^{p_g} \left\{ \prod_{j=1}^{p_g} \lambda_{gj}^2 \right\}^{-b_g-1} (b_g^2)^{-\frac{3}{2}} \exp \left\{ -\frac{1}{e_g b_g^2} \right\}$$

$$p(e_g | \cdot) \propto (e_g)^{-2} \exp \left\{ -\frac{1}{e_g} \left(1 + \frac{1}{b_g^2} \right) \right\}$$

Appendix C Scale-mixture representation of distribution

This section discusses the scale-mixture representation of following distributions :

Half cauchy distribution : $X^2 | v \sim \text{IG}(\frac{1}{2}, \frac{1}{v})$ and $v \sim \text{IG}(\frac{1}{2}, 1)$ implies $X \sim \mathcal{C}^+(0, 1)$.

Beta prime distribution : $X^2 | v \sim \text{IG}(b, \frac{1}{v})$ and $v \sim \text{IG}(a, 1)$ implies $X \sim \mathcal{B}'(a, b)$.

Appendix D Confusion matrix and performance measures

A confusion matrix summarizes the performance of a binary classifier:

		Actual Class	
		Positive (1)	Negative (0)
Predicted Class	Positive (1)	True Positive (TP)	False Positive (FP)
	Negative (0)	False Negative (FN)	True Negative (TN)

Table 11: Structure of a Binary Confusion Matrix

Measure	Formula	Interpretation
Sensitivity (Recall)	$\frac{TP}{TP + FN}$	Proportion of actual positives correctly identified
Specificity	$\frac{TN}{TN + FP}$	Proportion of actual negatives correctly identified
Precision	$\frac{TP}{TP + FP}$	Proportion of predicted positives that are actually positive
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall proportion of correct predictions
F1-Score	$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of precision and recall

Table 12: Common Performance Measures

Appendix E Gibbs sampler implementations in R

The R functions implementing the Gibbs samplers for the three methods are provided in this appendix. In addition, the GRASP and GIGG implementations make use of several auxiliary functions, prefixed with `mh_update_`, to carry out the MHwG updates. For brevity, these auxiliary functions are not reproduced here.

E.1 Gibbs sampler for BGHS

```
1 bghs_gibbs = function(data, nsamples = 10000, burnin = 1000,
2                       init, group_sizes, print_every = 100){
3
4   n = nrow(data); p = ncol(data) - 1; X = as.matrix(data[, -1]); y =
5     as.vector(data[, 1])
6   G = length(group_sizes); expected_len = n + 3*p + 2*G + 2
7
8   # Few preliminary checks
9   if(length(init) != expected_len) stop("init length does not match number of
10     parameters.")
11   if(burnin < 0) stop("burnin must be non-negative.")
12   if(burnin >= nsamples) stop("burnin must be smaller nsamples.")
13   if(sum(group_sizes) != p) stop("sum(group_sizes) must equal no. of
14     covariates.")
15
16   # Creating the sample dataframe
17   colname = c(paste0("w_", 1:n), paste0("beta_", 1:p),
18     unlist(Map(function(i, n) paste0("lam_sq_", i, 1:n), 1:G,
19       group_sizes)),
20     unlist(Map(function(i, n) paste0("t_", i, 1:n), 1:G, group_sizes)),
21     paste0("delta_sq_", 1:G),
22     paste0("c_", 1:G),
23     "tau_sq", "v"
24   )
25   sample = matrix(NA, nrow = nsamples, ncol = length(colname))
26   colnames(sample) = colname
27
28   # Tweaking initials
29   current_w = init[1:n]; current_beta = init[(n+1) : (n+p)]
30   current_lambda_sq = init[(n+p+1) : (n+2*p)]; current_t = init[(n+2*p+1) :
31     (n+3*p)]
32   current_delta_sq = init[(n+3*p+1) : (n+3*p+G)]; current_c =
33     init[(n+3*p+G+1) : (n+3*p+2*G)]
34   current_tau_sq = init[n+3*p+2*G+1]; current_v = init[n+3*p+2*G+2]
35
36   # initial row
37   sample[1,] = init
38
39   # Few predefined things
40   k = y - 0.5
41   idx = split(1:length(current_beta), rep(1:G, times = group_sizes)) # for
42     generating delta square
43
44   # Time related (To keep track of how much work has been completed and how
45     much remains)
46   start_time = Sys.time()
```

```

39 cat("\n=====\\n")
40 cat("          Starting BGHS Gibbs Sampler\\n")
41 cat("=====\\n")
42 cat("\\n\\n")
43 cat("-----\\n")
44 cat("Summary of experimental setting\\n")
45 cat("-----\\n\\n")
46 cat(sprintf("Observations      : %d\\n", n))
47 cat(sprintf("Covariates        : %d\\n", p))
48 cat(sprintf("Number of groups: %d\\n", G))
49 cat(sprintf("Group sizes       : %s\\n", paste(group_sizes, collapse = ", ")))
50
51 cat("\\n-----\\n\\n")
52
53 for(i in 2:nsamples){
54
55   # progress printing
56   if(i %% print_every == 0){
57     elapsed = as.numeric(difftime(Sys.time(), start_time, units = "secs"))
58     avg_time = elapsed/(i-1)
59     eta_remaining = avg_time * (nsamples - i)
60     pct = round(100*i/nsamples,1)
61     current_time = format(Sys.time(), "%H:%M:%S")
62
63     cat(sprintf(
64       "Time: %s => Samples: [%d/%d] | Completed: %.1f%% | elapsed: %.1fs |
65       ETA: %.1fs\\n",
66       current_time, i, nsamples, pct, elapsed, eta_remaining
67     ))
68   }
69
70   # Sample w
71   eta = as.vector(X %*% current_beta)
72   current_w = BayesLogit::rpg(num= n, h = 1, z = eta)
73
74   # Sample beta
75   XtOmegaX = crossprod(X, X*current_w)
76   D_inv = diag(1 / (rep(current_delta_sq, times = group_sizes) *
77     current_lambda_sq * current_tau_sq))
78   A = XtOmegaX + D_inv
79   R = chol(A)
80   Sigma_beta = chol2inv(R)
81   mu_beta = Sigma_beta %*% crossprod(X, k)
82   current_beta = MASS::mvrnorm(n = 1, mu = as.vector(mu_beta),
83     Sigma = Sigma_beta)
84
85   # Sample lambda square
86   current_lambda_sq = MCMCpack::rinvgamma(p, shape = 1, scale =
87     (current_beta^2)/(2*current_tau_sq*rep(current_delta_sq,
88     group_sizes)) + (1/current_t))
89
90   # Sample t
91   current_t = MCMCpack::rinvgamma(p, shape = 1, scale = 1 +
92     (1/current_lambda_sq))
93
94   # Sample delta square
95   current_delta_sq = sapply(1:G, function(g){

```

```

91     ind = idx[[g]]
92     shape = (group_sizes[g] + 1)/2
93     scale = (1/(2*current_tau_sq))*sum(current_beta[ind]^2 /
94         current_lambda_sq[ind]) + 1/current_c[g]
95     MCMCpack::rinvgamma(1, shape = shape, scale = scale)
96 })
97
98 # Sample c
99 current_c = MCMCpack::rinvgamma(G, shape = 1, scale = 1 +
100     (1/current_delta_sq))
101
102 # Sample tau
103 current_tau_sq = MCMCpack::rinvgamma(1,
104     shape = (p+1)/2,
105     scale =
106         0.5*sum(current_beta^2/
107             (current_lambda_sq *
108                 rep(current_delta_sq,
109                     times =
110                         group_sizes)))
111             +
112             (1/current_v))
113
114 # Sample v
115 current_v = MCMCpack::rinvgamma(1, shape = 1, scale = 1 +
116     (1/current_tau_sq))
117
118 new_row = c(current_w, current_beta, current_lambda_sq, current_t,
119     current_delta_sq, current_c, current_tau_sq, current_v)
120 sample[i,] = new_row # adding new row
121 }
122
123 # Excluding the samples of burn-in periods
124 sample = as.data.frame(sample)
125 post_idx = seq(burnin + 1, nsamples, by = 1)
126 posterior_sample = sample[post_idx,]
127 rownames(posterior_sample) =seq_len(nrow(posterior_sample))
128
129 # Time related (End)
130 elapsed =as.numeric(difftime(Sys.time(),start_time,units = "secs"))
131 cat("\n-----\n")
132 cat("Run Summary\n")
133 cat("-----\n")
134 cat(sprintf("Total samples : %d\n",nsamples))
135 cat(sprintf("Burn-in : %d\n",burnin))
136 cat(sprintf("Posterior kept: %d\n",nrow(posterior_sample)))
137 elapsed_min = floor(elapsed / 60)
138 elapsed_sec = round(elapsed %% 60)
139 cat(sprintf("Elapsed time : %.1f seconds (%dm %02ds)\n",elapsed,
140     elapsed_min, elapsed_sec))
141 cat("-----\n\n")
142 return(posterior_sample)
143 }

```

Listing 1: R code for the Gibbs sampler under BGHS

E.2 Gibbs sampler for GRASP

```

1 grasp_gibbs = function(data, nsamples = 10000, burnin = 1000,
2                       init, group_sizes, print_every = 100,
3                       sigma_prop_a = 0.2, sigma_prop_b = 0.2,
4                       sigma_prop_a_global = 0.2, sigma_prop_b_global = 0.2){
5
6   require(MASS)
7   require(BayesLogit)
8   require(MCMCpack)
9
10  n = nrow(data); p = ncol(data) - 1; X = as.matrix(data[, -1]);
11  y = as.vector(data[, 1]); G = length(group_sizes); expected_len = n + 3*p +
12    2*G + 2 + 4*G + 4
13
14  if(length(init) != expected_len) stop("init length does not match number of
15    parameters.")
16  if(burnin < 0) stop("burnin must be non-negative.")
17  if(burnin >= nsamples) stop("burnin must be smaller than nsamples.")
18  if(sum(group_sizes) != p) stop("sum(group_sizes) must equal number of
19    covariates.")
20
21  # Creating the sample dataframe
22  colname = c(paste0("w_", 1:n), paste0("beta_", 1:p),
23    unlist(Map(function(i, n) paste0("lam_sq_", i, 1:n), 1:G,
24      group_sizes)),
25    unlist(Map(function(i, n) paste0("t_", i, 1:n), 1:G,
26      group_sizes)),
27    paste0("delta_sq_", 1:G),
28    paste0("c_", 1:G),
29    "tau_sq", "v",
30    paste0("a_sq_", 1:G),
31    paste0("d_", 1:G),
32    paste0("b_sq_", 1:G),
33    paste0("e_", 1:G),
34    "a_sq", "d", "b_sq", "e"
35  )
36
37  sample = matrix(NA, nrow = nsamples, ncol = length(colname))
38  colnames(sample) = colname
39
40  # Tweaking initials
41  idx = 0
42  current_w = init[(idx+1):(idx+n)]; idx = idx + n
43  current_beta = init[(idx+1):(idx+p)]; idx = idx + p
44  current_lambda_sq = init[(idx+1):(idx+p)]; idx = idx + p
45  current_t = init[(idx+1):(idx+p)]; idx = idx + p
46  current_delta_sq = init[(idx+1):(idx+G)]; idx = idx + G
47  current_c = init[(idx+1):(idx+G)]; idx = idx + G
48  current_tau_sq = init[idx+1]; idx = idx + 1
49  current_v = init[idx+1]; idx = idx + 1
50  current_a_g_sq = init[(idx+1):(idx+G)]; idx = idx + G
51  current_d_g = init[(idx+1):(idx+G)]; idx = idx + G
52  current_b_g_sq = init[(idx+1):(idx+G)]; idx = idx + G
53  current_e_g = init[(idx+1):(idx+G)]; idx = idx + G
54  current_a_sq_global = init[idx+1]; idx = idx + 1
55  current_d_global = init[idx+1]; idx = idx + 1

```

```

51 current_b_sq_global = init[idx+1]; idx = idx + 1
52 current_e_global = init[idx+1]
53
54 # Few predefined things
55 idx_groups = split(1:p, rep(1:G, times = group_sizes))
56 k = y - 0.5
57
58 # initial row
59 sample[1,] = c(current_w, current_beta, current_lambda_sq, current_t,
60               current_delta_sq, current_c, current_tau_sq, current_v,
61               current_a_g_sq, current_d_g, current_b_g_sq, current_e_g,
62               current_a_sq_global, current_d_global, current_b_sq_global,
63               current_e_global)
64
65 # Time related (Initial)
66 start_time = Sys.time()
67 cat("\n=====\\n")
68 cat("          Starting GRASP Gibbs Sampler\\n")
69 cat("=====\\n\\n")
70 cat("-----\\n")
71 cat("Summary of experimental setting\\n")
72 cat("-----\\n")
73 cat(sprintf("Observations      : %d\\n", n))
74 cat(sprintf("Covariates          : %d\\n", p))
75 cat(sprintf("Number of groups: %d\\n", G))
76 cat(sprintf("Group sizes          : %s\\n", paste(group_sizes, collapse = ", ")))
77 cat("\\n-----\\n\\n")
78
79 for(iter in 2:nsamples){
80
81   # Progress Printing
82   if(iter %% print_every == 0){
83     elapsed = as.numeric(difftime(Sys.time(), start_time, units = "secs"))
84     avg_time = elapsed/(iter-1)
85     eta_remaining = avg_time * (nsamples - iter)
86     pct = round(100 * iter/nsamples,1)
87     current_time = format(Sys.time(), "%H:%M:%S")
88     cat(sprintf("Time: %s => Samples: [%d/%d] | Completed: %.1f%% |
89               elapsed: %.1fs | ETA: %.1fs\\n",
90               current_time, iter, nsamples, pct, elapsed, eta_remaining))
91   }
92
93   # Sample w
94   eta = as.vector(X %*% current_beta)
95   current_w = BayesLogit::rpg(num = n, h = 1, z = eta)
96
97   # Sample beta
98   D_vec = current_tau_sq * current_lambda_sq * rep(current_delta_sq, times
99             = group_sizes)
100   D_inv = diag(1/D_vec)
101   XtOmegaX = crossprod(X, X * current_w)
102   P = XtOmegaX + D_inv
103   R = chol(P)
104   mu_beta = backsolve(R, forwardsolve(t(R), crossprod(X,k)))
105   z = rnorm(p)
106   current_beta = mu_beta + backsolve(R, z)

```

```

106  # Sample lambda
107  b_g_vec = sqrt(current_b_g_sq[rep(1:G, times = group_sizes)])
108  delta_g_vec = rep(current_delta_sq, times = group_sizes)
109  scale_lambda = (current_beta^2) / (2*current_tau_sq*delta_g_vec) +
110                1/current_t
111  current_lambda_sq = rinvgamma(p, shape = b_g_vec + 1/2, scale =
112                             scale_lambda)
113
114  # Sample t
115  a_g_vec = sqrt(current_a_g_sq[rep(1:G, times = group_sizes)])
116  shape_t = a_g_vec + b_g_vec
117  scale_t = 1+(1/(current_lambda_sq))
118  current_t = rinvgamma(p, shape = shape_t, scale = scale_t)
119
120  # Sample delta_sq
121  b_global = sqrt(current_b_sq_global)
122  new_delta_sq = numeric(G)
123  for (g in 1:G) {
124    ind = idx_groups[[g]]
125    shape_delta = b_global + group_sizes[g] / 2
126    sum_term = sum(current_beta[ind]^2 / current_lambda_sq[ind])
127    scale_delta = sum_term / (2 * current_tau_sq) + 1 / current_c[g]
128    new_delta_sq[g] = rinvgamma(1, shape = shape_delta, scale = scale_delta)
129  }
130  current_delta_sq = new_delta_sq
131
132  # Sample c
133  a_global = sqrt(current_a_sq_global)
134  shape_c = a_global + b_global
135  scale_c = 1 + (1/current_delta_sq)
136  current_c = rinvgamma(G, shape = shape_c, scale = scale_c)
137
138  # Sample tau_Sq and v
139  total_sum = sum(current_beta^2 / (current_lambda_sq *
140    rep(current_delta_sq, times = group_sizes)))
141  shape_tau = (p + 1) / 2
142  scale_tau = 0.5 * total_sum + 1 / current_v
143  current_tau_sq = rinvgamma(1, shape = shape_tau, scale = scale_tau)
144  current_v = rinvgamma(1, shape = 1, scale = 1 + 1 / current_tau_sq)
145
146  # Sample auxiliary variables for hyperpriors (conjugate)
147  current_d_g <- rinvgamma(G, shape = 1, scale = 1 + 1 / current_a_g_sq)
148  current_e_g <- rinvgamma(G, shape = 1, scale = 1 + 1 / current_b_g_sq)
149  current_d_global <- rinvgamma(1, shape = 1, scale = 1 + 1 /
150    current_a_sq_global)
151  current_e_global <- rinvgamma(1, shape = 1, scale = 1 + 1 /
152    current_b_sq_global)
153
154  # Metropolis-Hastings updates for squared hyperparameters
155  # a_g_sq (per group)
156  for (g in 1:G) {
157    ind <- idx_groups[[g]]
158    t_vals <- current_t[ind]
159    current_a_g_sq[g] <- mh_update_a_g_sq_grasp(current_a_g_sq[g], t_vals,
160                                              current_d_g[g], group_sizes[g],
161                                              sigma_prop_a)

```

```

157 }
158 # b_g_sq (per group)
159 for (g in 1:G) {
160   ind <- idx_groups[[g]]
161   lambda_vals <- current_lambda_sq[ind]
162   t_vals <- current_t[ind]
163   current_b_g_sq[g] <- mh_update_b_g_sq_grasp(current_b_g_sq[g],
164     lambda_vals, t_vals,
165     current_e_g[g], group_sizes[g],
166     sigma_prop_b)
167 }
168 # a_sq_global
169 current_a_sq_global <- mh_update_a_sq_grasp(current_a_sq_global,
170   current_c,
171   current_d_global, G,
172   sigma_prop_a_global)
173 # b_sq_global
174 current_b_sq_global <- mh_update_b_sq_grasp(current_b_sq_global,
175   current_delta_sq, current_c,
176   current_e_global, G,
177   sigma_prop_b_global)
178
179 # Store samples
180 new_row <- c(current_w, current_beta, current_lambda_sq, current_t,
181   current_delta_sq, current_c, current_tau_sq, current_v,
182   current_a_g_sq, current_d_g, current_b_g_sq, current_e_g,
183   current_a_sq_global, current_d_global, current_b_sq_global,
184   current_e_global)
185 sample[iter, ] <- new_row
186 }
187
188 # Discard burn in and return
189 sample <- as.data.frame(sample)
190 post_idx <- seq(burnin + 1, nsamples, by = 1)
191 posterior_sample <- sample[post_idx, ]
192 rownames(posterior_sample) <- seq_len(nrow(posterior_sample))
193
194 elapsed <- as.numeric(difftime(Sys.time(), start_time, units = "secs"))
195 cat("\n-----\n")
196 cat("Run Summary\n")
197 cat("-----\n")
198 cat(sprintf("Total samples : %d\n", nsamples))
199 cat(sprintf("Burn-in : %d\n", burnin))
200 cat(sprintf("Posterior kept: %d\n", nrow(posterior_sample)))
201 elapsed_min <- floor(elapsed / 60)
202 elapsed_sec <- round(elapsed %% 60)
203 cat(sprintf("Elapsed time : %.1f seconds (%dm %02ds)\n", elapsed,
204   elapsed_min, elapsed_sec))
205 cat("-----\n\n")
206
207 return(posterior_sample)
208 }

```

Listing 2: R code for the Gibbs sampler under BGHS

E.3 Gibbs sampler for GIGG

```
1 gigg_gibbs = function(data, nsamples = 1000, burnin = 1000,
2                       init, group_sizes, print_every = 100,
3                       sigma_prop_a = 0.2, sigma_prop_b = 0.2){
4   require(MASS); require(BayesLogit); require(MCMCpack); require(GIGrvg)
5
6   n = nrow(data)
7   p = ncol(data) - 1
8   X = as.matrix(data[, -1])
9   y = as.vector(data[, 1])
10  G = length(group_sizes)
11  expected_len = n + p + p + G + 1 + 1 + 4*G
12  if(length(init) != expected_len) stop("init length does not match number of
13    parameters.")
14  if(burnin < 0) stop("burnin must be non-negative.")
15  if(burnin >= nsamples) stop("burnin must be smaller than nsamples.")
16  if(sum(group_sizes) != p) stop("sum(group_sizes) must equal number of
17    covariates.")
18
19  # creating the sample dataframe
20  colname = c(paste0("w_", 1:n), paste0("beta_", 1:p),
21             unlist(lapply(1:G, function(g) paste0("lam_sq_", g, "_",
22             1:group_sizes[g]))),
23             paste0("delta_sq_", 1:G), "tau_sq", "v",
24             paste0("a_sq_", 1:G), paste0("d_", 1:G),
25             paste0("b_sq_", 1:G), paste0("e_", 1:G))
26  sample = matrix(NA, nrow = nsamples, ncol = length(colname))
27  colnames(sample) = colname
28
29  # extract initial values
30  idx = 0
31  current_w = init[(idx+1):(idx+n)]; idx = idx + n
32  current_beta = init[(idx+1):(idx+p)]; idx = idx + p
33  current_lambda_sq = init[(idx+1):(idx+p)]; idx = idx + p
34  current_delta_sq = init[(idx+1):(idx+G)]; idx = idx + G
35  current_tau_sq = init[idx+1]; idx = idx + 1
36  current_v = init[idx+1]; idx = idx + 1
37  current_a_sq = init[(idx+1):(idx+G)]; idx = idx + G
38  current_d = init[(idx+1):(idx+G)]; idx = idx + G
39  current_b_sq = init[(idx+1):(idx+G)]; idx = idx + G
40  current_e = init[(idx+1):(idx+G)]; idx = idx + G
41  idx_groups = split(1:p, rep(1:G, times = group_sizes))
42  k = y - 0.5
43
44  # initial row
45  sample[1,] = c(current_w, current_beta, current_lambda_sq,
46                current_delta_sq, current_tau_sq, current_v, current_a_sq,
47                current_d, current_b_sq, current_e)
48
49  start_time = Sys.time()
50  cat("\n===== \n")
51  cat("          Starting GIGG Gibbs Sampler \n")
52  cat("===== \n \n")
53  cat("----- \n")
54  cat("Summary of experimental setting \n")
55  cat("----- \n")
56  cat(sprintf("Observations      : %d \n", n))
```



```

53 cat(sprintf("Covariates      : %d\n", p))
54 cat(sprintf("Number of groups: %d\n", G))
55 cat(sprintf("Group sizes    : %s\n", paste(group_sizes, collapse = ", ")))
56 cat("\n-----\n\n")
57
58 for(iter in 2:nsamples){
59   if(iter %% print_every == 0){
60     elapsed <- as.numeric(difftime(Sys.time(), start_time, units = "secs"))
61     avg_time <- elapsed/(iter-1)
62     eta_remaining <- avg_time * (nsamples - iter)
63     pct <- round(100 * iter/nsamples, 1)
64     current_time <- format(Sys.time(), "%H:%M:%S")
65     cat(sprintf("Time: %s => Samples: [%d/%d] | Completed: %.1f%% |
66               elapsed: %.1fs | ETA: %.1fs\n",
67                 current_time, iter, nsamples, pct, elapsed, eta_remaining))
68   }
69
70   # Sample w
71   eta = as.vector(X %*% current_beta)
72   current_w = BayesLogit::rpg(num = n, h = 1, z = eta)
73
74   # Sample beta
75   D_vec = current_tau_sq * current_delta_sq[rep(1:G, times = group_sizes)] *
76           current_lambda_sq
77   D_inv = diag(1/D_vec)
78   XtOmegaX = crossprod(X, X * current_w)
79   A = XtOmegaX + D_inv
80   R = chol(A)
81   Sigma_beta = chol2inv(R)
82   mu_beta = Sigma_beta%*% crossprod(X,k)
83   current_beta = MASS::mvrnorm(n = 1, mu = as.vector(mu_beta),
84                                Sigma = Sigma_beta)
85
86   # Sample lambda sq
87   b_g_vec = sqrt(current_b_sq[rep(1:G, times = group_sizes)])
88   delta_g_vec = rep(current_delta_sq, times = group_sizes)
89   shape_lambda = b_g_vec + 0.5
90   scale_lambda = 1 + (current_beta^2)/(2*current_tau_sq*delta_g_vec)
91   current_lambda_sq = rinvgamma(p, shape = shape_lambda, scale =
92                                 scale_lambda)
93
94   # Sample delta sq
95   new_delta_sq = numeric(G)
96   a_g_vec = sqrt(current_a_sq)
97   for(g in 1:G){
98     ind = idx_groups[[g]]
99     p_g = group_sizes[g]
100    sum_term = sum((current_beta[ind]^2)/(current_tau_sq *
101      current_lambda_sq[ind]))
102    lambda_gig = a_g_vec[g] - p_g/2
103    chi_gig = sum_term
104    psi_gig = 2.0
105    new_delta_sq[g] = rgig(1, lambda = lambda_gig, chi = chi_gig, psi =
106      psi_gig)
107  }
108  current_delta_sq = new_delta_sq
109
110  # Sample tau_sq and v

```

```

105     total_sum = sum((current_beta^2) /
106                     (current_lambda_sq * rep(current_delta_sq, times =
107                                             group_sizes)))
108     shape_tau = (p + 1)/2
109     scale_tau = 0.5*total_sum + 1/current_v
110     current_tau_sq = rinvgamma(1, shape = shape_tau, scale = scale_tau)
111     current_v = rinvgamma(1, shape = 1, scale = 1 + 1/current_tau_sq)
112
113     # Auxiliary variables
114     current_d = rinvgamma(G, shape = 1, scale = 1+1/current_a_sq)
115     current_e = rinvgamma(G, shape = 1, scale = 1+1/current_b_sq)
116
117     # Metropolis - Hatings update for a_sq and b_sq
118     # a_g^2 (per group)
119     for(g in 1:G){
120         current_a_sq[g] =
121             mh_update_a_g_sq_gigg(current_a_sq[g], current_delta_sq[g],
122                                   current_d[g], sigma_prop_a)
123     }
124     # b_g^2 (per group)
125     for(g in 1:G){
126         ind = idx_groups[[g]]
127         lambda_vals = current_lambda_sq[ind]
128         p_g = group_sizes[g]
129         current_b_sq[g] = mh_update_b_g_sq_gigg(current_b_sq[g], lambda_vals,
130                                                 current_e[g], p_g, sigma_prop_b)
131     }
132
133     # Store Samples
134     new_row = c(current_w, current_beta, current_lambda_sq,
135                current_delta_sq, current_tau_sq, current_v,
136                current_a_sq, current_d, current_b_sq, current_e)
137     sample[iter,] = new_row
138 }
139
140 # Discard burn in
141 sample = as.data.frame(sample)
142 post_idx = seq(burnin+1, nsamples, by = 1)
143 posterior_sample = sample[post_idx,]
144 rownames(posterior_sample) = seq_len(nrow(posterior_sample))
145
146 elapsed = as.numeric(difftime(Sys.time(), start_time, units = "secs"))
147 cat("\n-----\n")
148 cat("Run Summary\n")
149 cat("-----\n")
150 cat(sprintf("Total samples : %d\n", nsamples))
151 cat(sprintf("Burn-in       : %d\n", burnin))
152 cat(sprintf("Posterior kept: %d\n", nrow(posterior_sample)))
153 elapsed_min <- floor(elapsed / 60)
154 elapsed_sec <- round(elapsed %% 60)
155 cat(sprintf("Elapsed time  : %.1f seconds (%dm %02ds)\n", elapsed,
156             elapsed_min, elapsed_sec))
157 cat("-----\n\n")
158 return(posterior_sample)
159 }

```

Listing 3: R code for the Gibbs sampler under BGHS

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